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**CLICK TO VOTE: A DIGITAL VOTING SYSTEM FOR
A PUBLIC SENIOR HIGH SCHOOL**

A RESEARCH PAPER

Presented to
The Faculty of Senior High School
Iloilo National High School

In Partial Fulfillment of
the Requirements in Inquiries, Investigation, and Immersion

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ABSTRACT

This study aimed to design and develop "Click to Vote," a digital voting system for the Senior High School Department of Iloilo National High School, specifically towards the Iloilo National High School Senior High School Learner Government Election, to address the current problems found with manual validation and issues found when google forms was used during the last election.. Utilizing a descriptive-developmental research design, the system was constructed following the Software Development Life Cycle (SDLC) Waterfall and ADDE models to feature secure logins, user management, live election monitoring, election history and election statistics. The system was evaluated by 13 respondents, comprising 6 non-IT experts (Election Committee Officers) and 7 IT experts, using the ISO/IEC 25010:2011 Software Quality model and the System Usability Scale (SUS). The results revealed that Click to Vote achieved an overall software quality rating of "excellent" with a mean of 4.62. In terms of usability, both IT and Non-IT Experts interpreted it as "acceptable." The study concludes that Click to Vote is a viable and functional alternative to current practices specifically as a replacement to google forms in elections, though further enhancements are recommended to maximize long-term success and implementation especially with the new curriculum that is, as of writing, currently being implemented within Iloilo National High School.

Keywords: vote, elections, digital voting, senior high school





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CHAPTER I

INTRODUCTION OF THE STUDY

Chapter One, Introduction of the Study, consists of five (5) parts, namely: (1) Background of the Study, (2) Objectives of the Study, (3) Definition of Terms, (4) Significance of the Study, and (5) Scope and Delimitation of the Study.

Part One, Background of the Study, establishes the foundation of the research. justifies its relevance, and highlights the circumstances that led to the selection of the problem. It also describes the study's context and conceptual framework.

Part Two, Objectives, outlines the primary goals of the study it seeks to achieve.

Part Three, Definition of Terms, provides the conceptual and operational definitions of key terms and concepts used throughout the study to ensure clarity and consistency.

Part Four, Significance of the Study, discusses the potential contributions of the research and identifies the beneficiaries of its findings.

Part Five, Scope and Delimitation of the Study, defines the parameters of the study, specifying its focus, coverage, and limitations to provide a clear boundary for the investigation.



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Background of the Study

Election integrity is the foundation of democratic societies. According to the International Institute for Democracy and Electoral Assistance (n.d.), credible elections are essential in ensuring that the will of the people is accurately reflected, strengthening public trust in governance and democratic institutions (IDEA, 2012). In the Philippines, this principle is upheld through Batas Pambansa Blg. 881, also known as the Omnibus Election Code of the Philippines, which provides legal frameworks to ensure free, orderly, honest, peaceful, and credible elections. It establishes safeguards against electoral fraud, coercion, and misconduct. Election integrity ensures that elections are conducted fairly, transparently, and without interference. Norris (2014) mentions how voters, when confident in the electoral process, are more likely to participate and trust its outcomes. This confidence can be achieved when data privacy and security are ensured, which play a crucial role in upholding election integrity and preserving the democratic values upon which societies are built (Norris, 2014).



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Conventional elections in the contemporary era have increasingly been perceived as inefficient and vulnerable to security risks. For instance, Norris (2014) found that traditional elections are often time-consuming, prone to human error, and susceptible to electoral malpractices such as ballot tampering, vote buying, and delays in the canvassing of results. Similarly, the ACE Electoral Knowledge Network (n.d.) and the International Institute for Democracy and Electoral Assistance (IDEA, n.d.) emphasized that manual voting processes, weak voter authentication, lack of transparency in vote counting, and physical interference at polling stations significantly contribute to the vulnerability of conventional elections. These weaknesses run contrary to Republic Act No. 7166 and Republic Act No. 9369, which mandate clean, honest, accurate, meaningful, and credible elections, as well as to the principles enshrined in the 1987 Philippine Constitution, which upholds the sanctity of the ballot and the people’s right to suffrage.

In response to these challenges, digital elections have emerged as an alternative electoral system that utilizes modern technologies such as the internet, computers, and electronic devices to improve efficiency and security in the voting process (Wadowski,



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2024). The concept of digital elections was introduced to reduce human intervention, speed up vote tallying and transmission of results, enhance transparency, and strengthen election security through mechanisms such as encryption, automated verification, and access control systems. As advancements in information and communication technology continue, digital voting has been increasingly explored and implemented in various countries and institutions as a means of modernizing electoral processes while maintaining election integrity and public trust. Robust security measures are logical solutions. They protect the ideology and are the cornerstone of electronic voting systems, protecting against fraud, tampering, and unauthorized access, through advanced encryption protocols, multi-factor authentication, secure infrastructure, audit trails, tamper-resistant hardware and software, continuous monitoring, independent assessments, and collaborations, electronic voting strives to ensure election integrity. By upholding the highest standards of security, it plays a vital role in preserving trust in the democratic process.



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The 2025-2026 Iloilo National High School - Senior High School Learners'

Government (INHS-SHSLG) digital election prioritized security and integrity, earning a high mean rating of 3.80. Yet, the process exceeded expected timelines due to these rigorous measures: a temporal delay noted by the Senior High School Election Committee adviser that remains underexplored beyond standard security practices (Bugayong et al., 2025). This gap highlights the need for a more efficient method of conducting elections to save time and effort on the parts of the election committee and the gap between the election proper and the posting of results. Hence the objectives of the study:



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Objectives

The study aims to develop Click to Vote, a digital voting system for a Public Senior High School.

Specifically, this study aims to:

1. develop an online voting system for the SSLG-SHS election which performs the following functions:
 - 1.1 online voting, monitoring and auditing of elections,
 - 1.2 tracking of statistics like number of voters and number of voters per grade level, track/cluster and section
2. determine the overall quality of the developed system based on the ISO/IEC 25010:2011 Systems and Software Quality Requirements and Evaluation (SQuaRE) Model as evaluated by the IT experts;
3. determine the system usability of the developed system using the System Usability Scale (SUS) as evaluated by the IT experts and end users.



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Significance of the Study

The researcher embarked on this study for the very reason that the results would significantly contribute to existing knowledge. Significantly, the findings would be beneficial to the following:

Learners. The system provides a convenient and secure platform for casting votes, reducing time and effort compared to traditional manual voting. It encourages higher participation and fosters a sense of responsibility in democratic processes.

Student Council. The system ensures fair and transparent elections, strengthening the legitimacy of elected officers. It also streamlines the voting process, allowing the council to focus on leadership and service rather than manual election.

School Administrators and Election Committees. The system reduces manual workload, minimizes human error, and accelerates vote counting, enabling efficient and accurate election results. It also enhances accountability through automated data recording and reporting.

Teachers and Staff. By simplifying election management, the system allows educators to focus more on academic responsibilities rather than logistical tasks.



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Future Researchers. This study serves as a reference for developing similar systems in other educational institutions, contributing to the advancement of digital governance in schools.

Definition of Terms

Voter. A learner who is currently enrolled in their school has the right to cast their vote for their candidates during the election period (Department of Education, 2023). It refers to an INHS Senior High School student registered in the Click to Vote system who successfully logs in and submits an electronic ballot during the SHSLG election.

Senior High School Election Committee Adviser. A teacher who monitors the election activities, guides the process from candidacy filing to winner proclamation, and ensures compliance with guidelines (Department of Education, 2023). They assigned to oversee the election using the Click to Vote system, validate procedures, and approve final results generated by the system.

Senior High School Election Committee Officer. A student volunteers who forms the election body in schools, handling tasks like screening certificates of candidacy, campaigning oversight, and result validation (Department of Education, 2023). They operate under adviser supervision to maintain transparency in school election polls especially in the SHSLG elections.



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SHSLG (Senior High School Learners Government). A student council for learners in the senior high school level (Department of Education, 2023). It promotes leadership through elections involving forums, campaigns, and voting. It refers specifically to the INHS Senior High School Learners’ Government for School Year 2025–2026, whose officers are elected through the Click to Vote digital voting system.

Digital Election. A mode of where they conduct the elections both casting and counting of votes virtually through online platforms (Department of Education, 2023). In this study, a digital election is the SHSLG election implemented via the Click to Vote website, where voters log into the system, select candidates on-screen, and submit ballots that are automatically stored, tallied, and displayed in real time to authorized administrators.



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Scope and Delimitation

This study aims to design Click to Vote: A Digital Voting System for Public Senior High Schools, which seeks to provide a more efficient, secure, and accessible alternative to traditional voting methods. This study is, however, limited to the Iloilo National High School's SHSLG election for School Year 2025 to 2026. The scope is confined due to the researchers' limited access to faculty members and administrative authorities, which restricted extensive coordination and validation of the system's processes.

The developed system will include features such as: Voting Online, Election Management, Automatic Tallying, Live Monitoring, Election Override and Audit logs. Certain features will be locked based on user roles which include: Admins, Advisers, Officers and Voters. Each with a corresponding access level which will determine what feature that role has access to. With admins having the highest access level of 3 and voters having the lowest at 0.

Additionally, the sample space was also limited to selected participants directly involved in the SHSLG election, thereby excluding the entire student population and other public senior high schools.



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CHAPTER II

REVIEW OF RELATED LITERATURE

Chapter 2 presents the review of related literature, studies and systems relevant to the present study. This chapter provides a foundation for understanding previous research, existing systems, and theoretical bases that support the development of the proposed system. It is organized into five (5) major parts: (1) Related Literature, (2) Related Systems, (3) Theoretical Framework, (4) Conceptual Framework, and (5) Synthesis.

Part One, Related Literature, discusses selected studies that define and explain election systems, highlighting their purposes, features, and importance in educational institutions

Part Two, Related Systems, presents existing digital election systems that are currently used by educational institutions, focusing on their functions and relevance to the present study.

Part Three, Theoretical Framework, explains the theories and models that guide the design, development, and evaluation of the election system.

Part Four, Conceptual Framework, illustrates the relationship among the key variables, concepts, and processes involved in the study through a visual paradigm.





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Part Five, Synthesis, summarizes and compares the reviewed literature and systems, identifies similarities and differences, and highlights the research gap that the present study aims to address.

Related Literature

Tabanao and Cece (2025) conducted a study on the development of an online voting application at Negros Oriental State University–Guihulngan. Their research focused on replacing manual election processes with a secure digital system incorporating authentication, encryption, and real-time monitoring to ensure vote integrity. Post-election surveys revealed that 95% of users preferred the system over traditional methods due to improved security, usability, and reliability. The findings support the scalability of secure online voting systems in educational institutions and validate iterative security enhancements within the Philippine educational context.

Bugayong et al. (2025) examined a public senior high school learner government election conducted digitally via Google Forms. The study found that the election was quantitatively rated as “very secure” and “very transparent.” However, despite positive results, election officers reported operational challenges, particularly the manual validation of voter information such as section assignments. This verification process took approximately seven hours—longer than the three-hour election



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period—highlighting inefficiencies in data validation mechanisms and emphasizing the need for automated verification systems.

Debant and Hirschi (2023) analyzed the French legislative online voting protocol and identified serious security vulnerabilities. Their research revealed flaws in receipt verification mechanisms that could allow compromised servers to alter ballots without detection. Additionally, weaknesses in the zero-knowledge proof implementation exposed risks to voter privacy and ballot redirection. Their findings stress the importance of robust cryptographic binding, secure client-side validation, and end-to-end verifiability in electronic voting systems.

Pugeda (2025) explored the impact of implementing an e-voting system on student governance, as detailed in The GLOW. The study emphasized how digitalization improves accessibility, transparency, and efficiency in student elections. Findings indicated increased voter turnout, real-time result tabulation, and improved eligibility verification through integration with student information systems. The research highlights the transformative role of e-voting in strengthening democratic participation within academic institutions.



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Related Systems

De los Santos et al. (2020) introduced an e-voting kiosk system designed for school-based registration and voting. The system utilized a networked architecture connecting kiosks for biometric authentication and encrypted ballot transmission. Implemented in a Philippine school setting, the system reduced voting time by 40% compared to manual elections while maintaining vote accuracy above 99%. This hardware-supported model demonstrates the feasibility of localized electronic voting infrastructures in educational institutions.

Estonia's national e-voting infrastructure (pollsID.ee, 2025) represents a mature, large-scale digital election system adaptable to smaller institutions. The system uses government-issued digital IDs (e-ID) for secure voter authentication and employs end-to-end encryption with audit mechanisms to ensure verifiability while preserving voter anonymity. Since 2005, it has processed millions of votes with fraud rates reported below 0.001%. Its standardized authentication framework serves as a model for secure digital election systems.

Tahir et al. (2025) developed an online voting platform tailored for student elections in a Malaysian university. The system incorporates multi-factor authentication, real-time vote tallying, and anomaly detection algorithms to prevent duplicate voting and



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cyber threats. During testing, it achieved a 95% voter turnout and successfully detected all simulated fraud attempts. The platform also featured public dashboards to enhance transparency. This software-based system highlights the effectiveness of integrity-focused digital election technologies in academic environments.



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Comparative Analysis

Table 1

Comparative Analysis of Click to Vote vs other systems

System	Security	Integrity	User Friendliness
Traditional Voting	Security measures rely on personnel to be present at the voting site	Results and Processes are kept hidden with very few occasionally showing transparency	Simple concept, very easy to follow and use.
E-Voting Kiosk	Security follows similarly to traditional voting where selected machines have the program installed which is attended by personnel.	Showcases current updates on the election results in real time and actively shows its processes.	Very simple and easy to vote in. Can see live updates on the current state of the election.
Estonia E-Voting	Requires voters to use an approved device and a certified application alongside 2 different PINs in order to vote. 1 for logging in and another for submitting the vote	Processes are not revealed to the public and results are only shown at the end of the election	Alright but quite lengthy, voting itself requires the second PIN which may be a hassle and there's a section about knowing the candidates first before voting which may be a hindrance to those who already know about the candidates before hand
Click to Vote	Uses a list of enrolled learners provided by an external application which contains information needed for login that can only be found in said third party system or the learners registration form.	Reveals the current state of the election, number of voters, candidates, time left and how many votes per candidate	Easy and simple for actual voting, but quite lengthy login process requiring 4 fields to be input.



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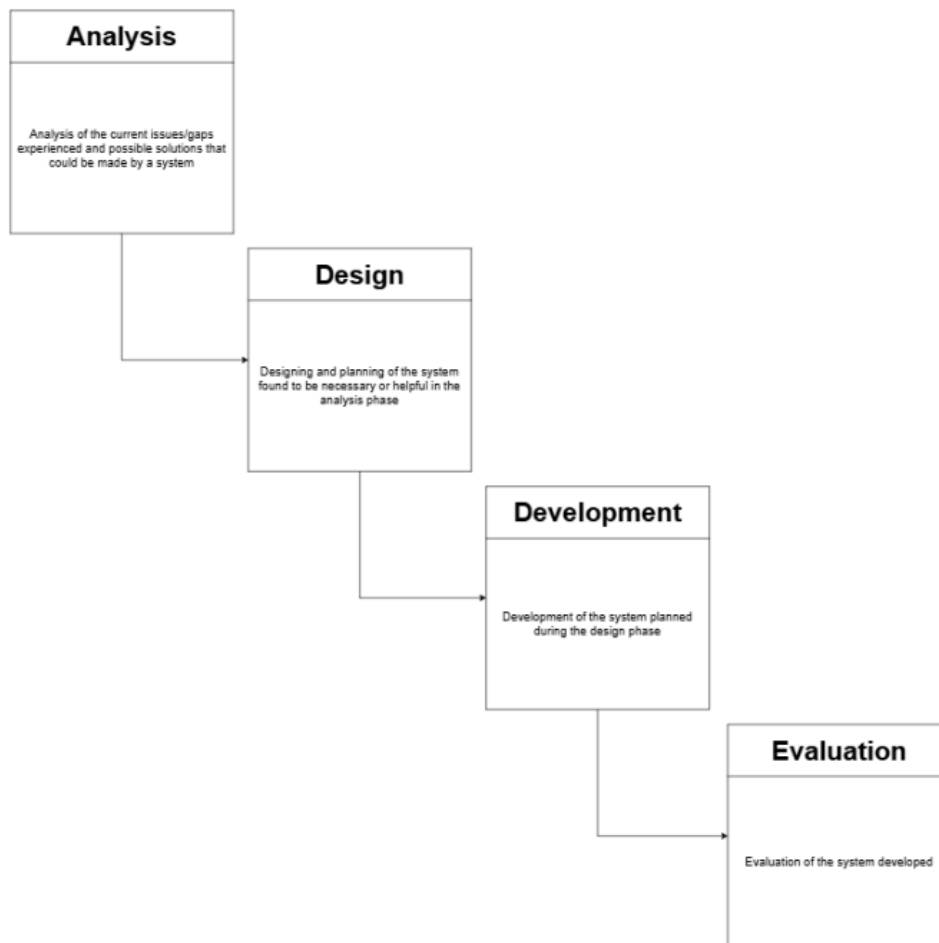
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Theoretical Framework

This study will make use of 3 theories: the Software Development Life Cycle (SDLC), the Information Systems (IS) Success Model, and the Technological Acceptance Model (TAM).

Figure 1

The Modified Waterfall Model





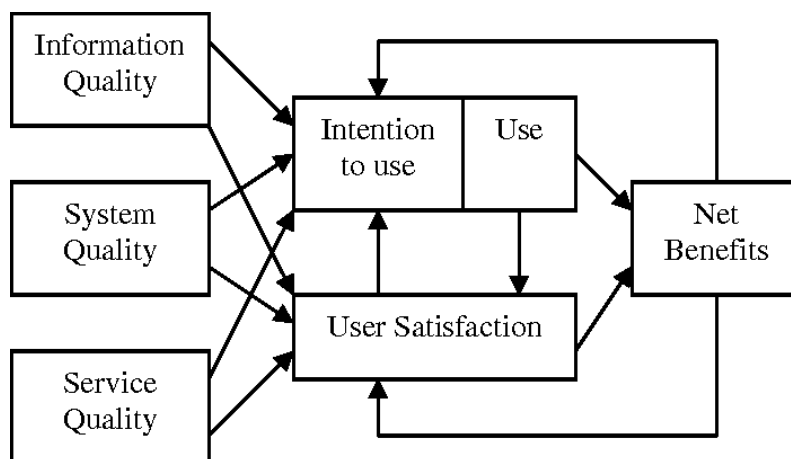
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To begin, the Software Development Life Cycle is a framework that defines the steps involved in the software development process at each stage. It provides a detailed plan for developing, deploying, and maintaining the software (Vijay, 2025). SDLC methodologies could guide the systematic development of software, ensuring that election systems are reliable goals. The most common methodology is the waterfall model. Rahmayani Early & Alhuda (2022) and Hafiz et al. (2020) label it as something applicable and highly recommended for an election system. According to Tutorialspoint (n.d.), the sequential phases of this model include requirement analysis, system design, implementation, testing, deployment, and maintenance.

Figure 2

The Information Systems (IS) Success Model.



The Information Systems (IS) Success Model is another useful framework for assessing the effectiveness of information systems (Mahmoud et al., 2025).

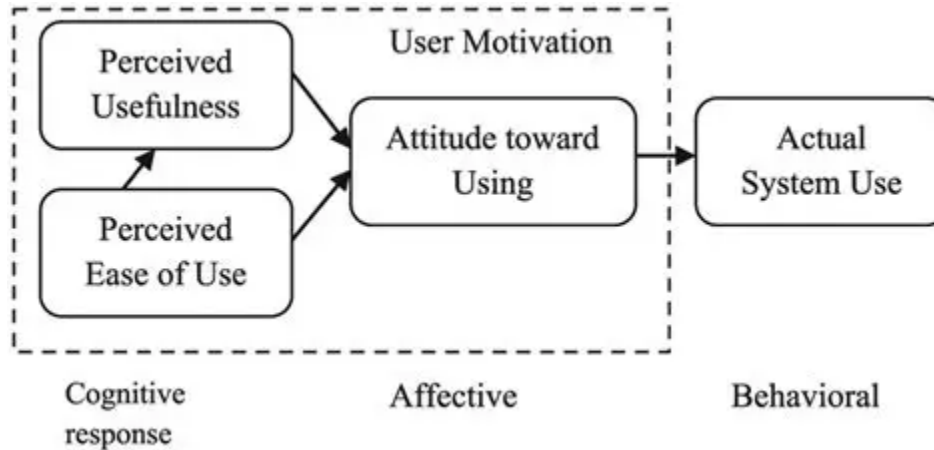


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Figure 3

The Technology Acceptance Model



Lastly, Granić's (2025) instructive material cites Fred Davis' Technology Acceptance Model (1986), which depicts user pleasure, accessibility, and usability. The model proposes that, in a theoretical situation, the characteristics of a technological system influence potential users' incentive to adopt it. This motivation is primarily impacted by two key factors: perceived ease of use and perceived usefulness, which tie back to the evaluation of Click to Vote by SHS Election Committee and IT experts. The online system would be rated by means of usability of both UX (user experience) and UI (user interface). Although implementation and analysis of behavioral intention when using the system in this study is not possible, the Technology Acceptance Model can be used as a basis for user satisfaction and inquiry for engagement (Suriani, 2025).



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Conceptual Framework

Figure 4

Conceptual Framework of Click to Vote

Input	Key Stakeholder Feedback/Suggestions - SHS Election Committee Adviser via interviews - SHS Election Committee Officers via group discussion - IT Expert Comments Issues Currently in the most recent Online Election (via Google Forms) - Voters inputting their information manually, introducing human error - Manual rechecking and validation taking too much time - Google forms not being dynamic/adaptive depending on the voter's information - Too many google forms specialized for each Track/Cluster because of the Board Member position Data Requirements - All Users - Name, Email, Password, UID, Role, Access Level - Voters and Candidates - Grade Level, Track/Strand/Cluster, Section - Elections - Election Schedule, Assigned Users, Audits - External CSV Provided by an External System Evaluation Tools - ISO/IEC Software Quality and Requirements Evaluation (SQuaRE) Model - System Usability Scale (SUS)
Process	System Design - Flow of an election mimicking an actual election (see comparison table in synthesis for details) - Database structured in a way that follows normalization and automatically validating and counting votes - UI displaying a clear flow from start to end of an election as well as statistics of current and ongoing elections System Development - Front-end - Login, Election Dashboard, History, Voters, Officers, Candidates, Audits, Voting Screen - Back-end - Validation, Success and Error messages, Error Catches, Prevent user input errors Testing and Evaluation - Pilot Testing and Prototype Showcase to SHS Election Committee Officers - IT and Non-IT Expert Evaluation - IT Expert comments and suggestions - Implementation of recommended features - Fixing of issues found
Output	Developed and Tested - Click to Vote - Evaluation Results

The conceptual framework for Click to Vote is based off of the Input, Process, Output (IPO) model. Each of the 3 sections of the framework are divided into 2 parts, the top part for “development” and the bottom for “operation.” This study will concern itself over the “development” part, but the system will rely on the “operation” part of the framework.



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Input. The framework highlights the need for interviews with the clientele in order to gain a baseline for how the system will function. Afterwards, continual feedback from them as well as the system gets developed to ensure the system remains within the scope of the study and doesn't go too far with ambitions. Or features that may become impossible to develop.

Process. After the inputs from the clientele, development of the many diagrams for the system was started. These diagrams reflect how the system will work in a visual format to ensure that development stays on target and keep track of what has been, and what needs to be developed.

Output. And finally, once the system has been developed according to the Input and Processes, evaluation and more importantly, the results from those evaluations will then become the output for this study.



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Synthesis

The chapter emphasizes the efficiency of digital/electronic voting when compared to traditional methods. It discusses the potential for digital elections and security and integrity measures to ensure that the election remains fair for all. However, possible flaws in those security and integrity measures still exist and if there is a possibility for human error in the system, for instance, voters inputting their own information, would require the officers managing the election to manually check if the information inputted was correct. This shows that a digital voting system has potential to be more efficient, secure and transparent than a traditional election as long as the security measures are uncompromised, trusted by the public, and the system itself already has the voters' information for the officers to not need to manually check afterwards.



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CHAPTER III

METHODOLOGY

This chapter presents the research methods and procedures used in the design, development, and evaluation of Click to Vote: a digital voting system for a public senior high school. It describes the Purpose of the study, research design, research locale, system design, respondents, research instruments, data-gathering procedures, and data analysis techniques. The methodology is aligned with the requirements of the Inquiry, Investigation, and Immersion (III) course, which emphasizes systematic inquiry, ethical research practices, and evidence-based conclusions.

Purpose of the Study and Research Design

The purpose of this study was to develop Click to Vote. A system which will make online elections in INHS specifically for the SHSLG elections more convenient and more efficient in terms of preparation, actually conducting the election and post validation. The system was evaluated in terms of overall system quality and usability using ISO/IEC SQuaRE Model and the SUS respectively.





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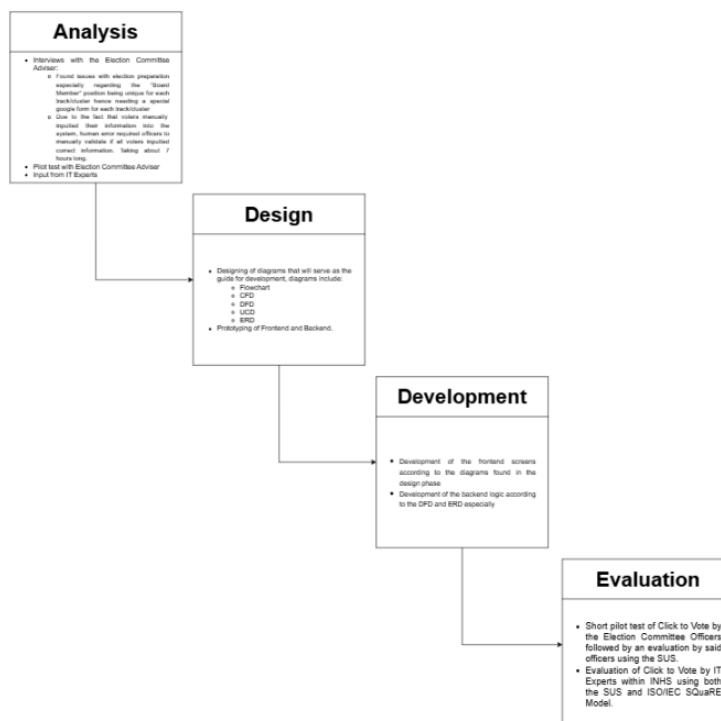
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The research study will follow a developmental research design, which, according to Ibrahim (2016), is an interactive, cyclical process of development and research in which the theoretical ideas of the designer inform the development of quality products. It is also a systematic study of designing, developing, and carefully evaluating instructional programs, processes, and products that must meet imposed mandates from a key stakeholder (Internet Public Library, 2021).

Methods

Figure 5

Modified Waterfall model of Click to Vote





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The system, Click to Vote, was designed following the Software Development Life Cycle (SDLC) using the revised Waterfall Model. This framework defines the steps involved in the software development process at each stage. It provides a detailed plan for developing, deploying, and maintaining the software (Vijay, 2025). SDLC methodologies could guide the systematic development of software, ensuring that election systems are reliable goals. The most common methodology is the waterfall model. Rahmayani Early & Alhuda (2022) and Hafiz et al. (2020) label it as something applicable and highly recommended for an election system. According to Tutorialspoint (n.d.), the sequential phases of this model include requirement analysis, system design, implementation, testing, deployment, and maintenance.



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Analysis

To analyze the research problem within Iloilo National High School, the researchers interviewed key stakeholders who might be involved in the development of Click to Vote, whether through physical advice or online messages. First was the SHS Elections Committee Adviser, who originally provided feedback in the Background of the Study, Outlining the problem of time when it comes to the validation of an election with the voters inputting their own information. Stating that the validation took longer than the election itself. They gave suggestions for the research scope and system GUI. Second were the actual SHS Elections Committee Officers, which supported their adviser's suggestions and added their own insights from having performed elections before.

Design

After analysis of the current issues experienced by the previous election using google forms, the researchers then used the information gathered during analysis to dictate the creation of necessary diagrams which will then guide the development of Click to Vote. These diagrams include: Context Flow Diagram, Data Flow Diagram, Use Case Diagram and Entity Relationship Diagram



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Context Flow Diagram

Figure 6

Context Flow Diagram.



The Context Flow Diagram (CFD) presents a view of the interaction between the system and its users. The CFD for Click to Vote illustrates what specific roles can do within the system. Voters are only allowed to login, vote and see election results once the election is done for their assigned elections. Officers are able to assign voters and candidates as well as see the information for their assigned election. Advisers, on top of having all the privileges of Officers, are also able to override, end or extend the voting time of their assigned election if they deem it necessary. And finally, Admins have all the privileges as Advisers but are able to manage all elections.



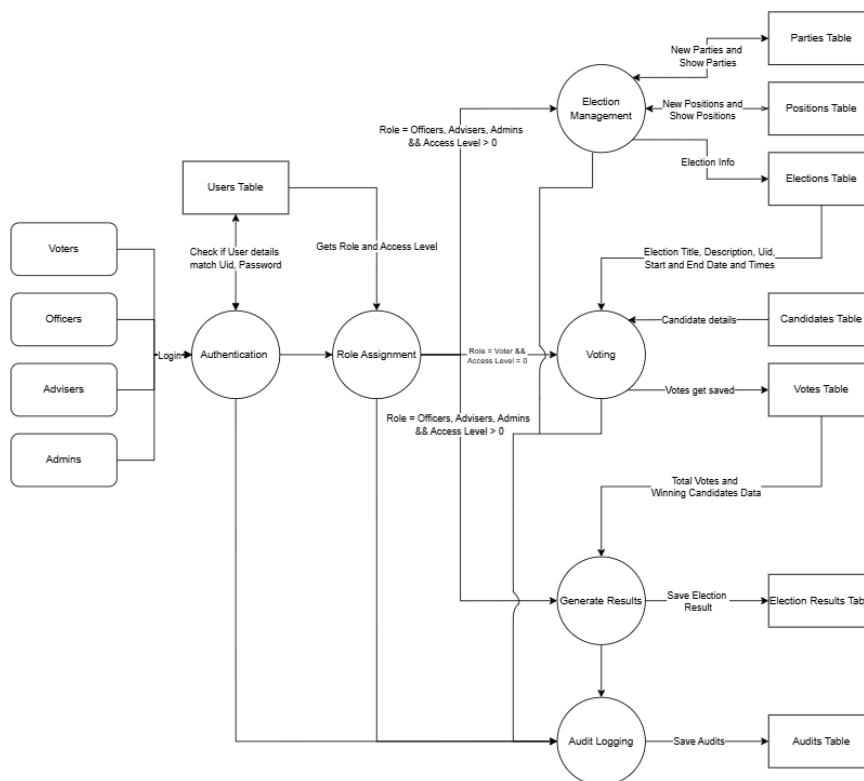
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Data Flow Diagram

Figure 7

Data Flow Diagram.



The Data Flow Diagram (DFD) illustrates how data moves within the system. It shows the process of data input, processing, storage, and output. DFDs are generally more in-depth versions of CFDs. Showcasing how data flows from the user and through the system to finally creating the desired output to the user which is exactly the case for Click to Vote.



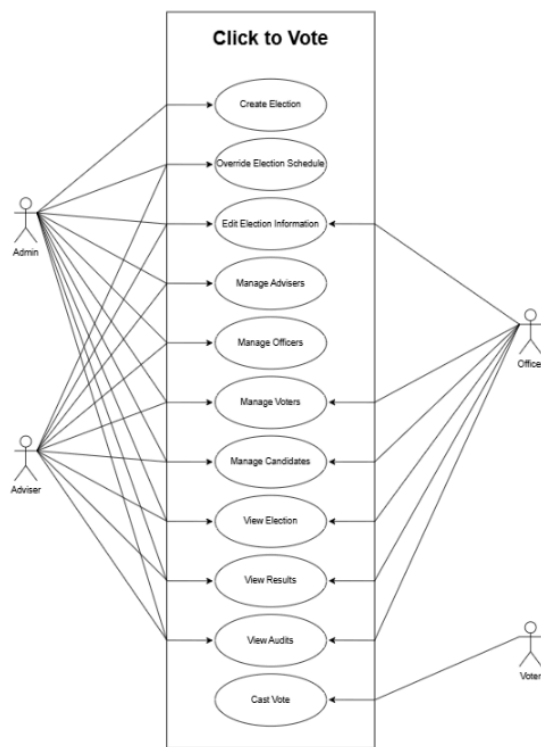
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Use Case Diagram

Figure 8

Use Case Diagram.



The Use Case Diagram (UCD) depicts the possible actions and interactions users can perform within the system. In the case of Click to Vote, the users include Admin, Adviser, Officer and Voter. While the uses cases include: Create Election, Override Election Schedule, Edit Election Information, Manage Advisers, Manage Officers, Manage Voters, Manage Candidates, View Election, View Results and Cast Vote.



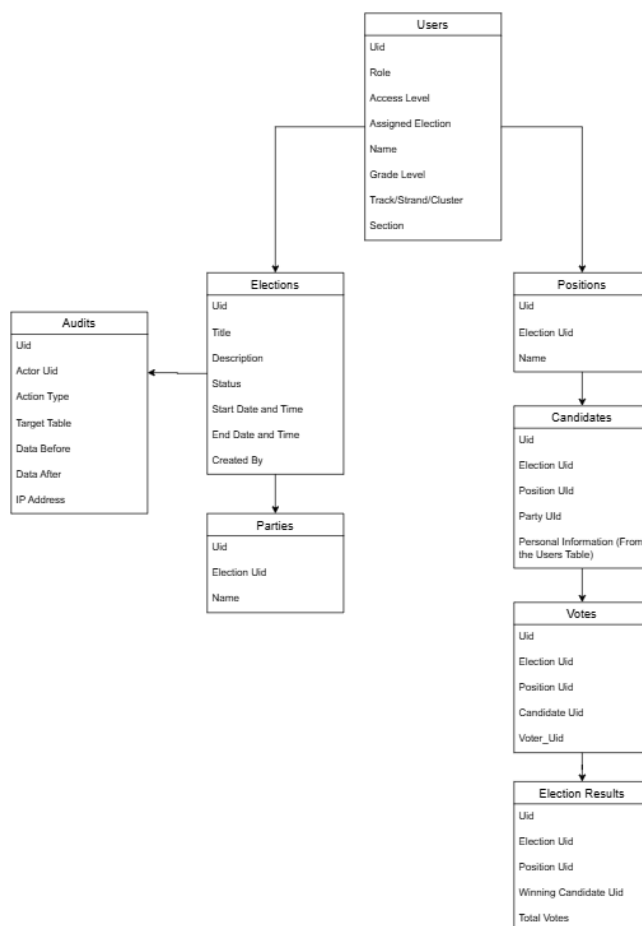
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Entity Relationship Diagram

Figure 9

Entity Relationship Diagram



The Entity Relationship Diagram (ERD) shows how “Entities” interact with each other in the system. Starting from the users, and through the database tables. This ERD depicts how data in the database flows and where it's stored.



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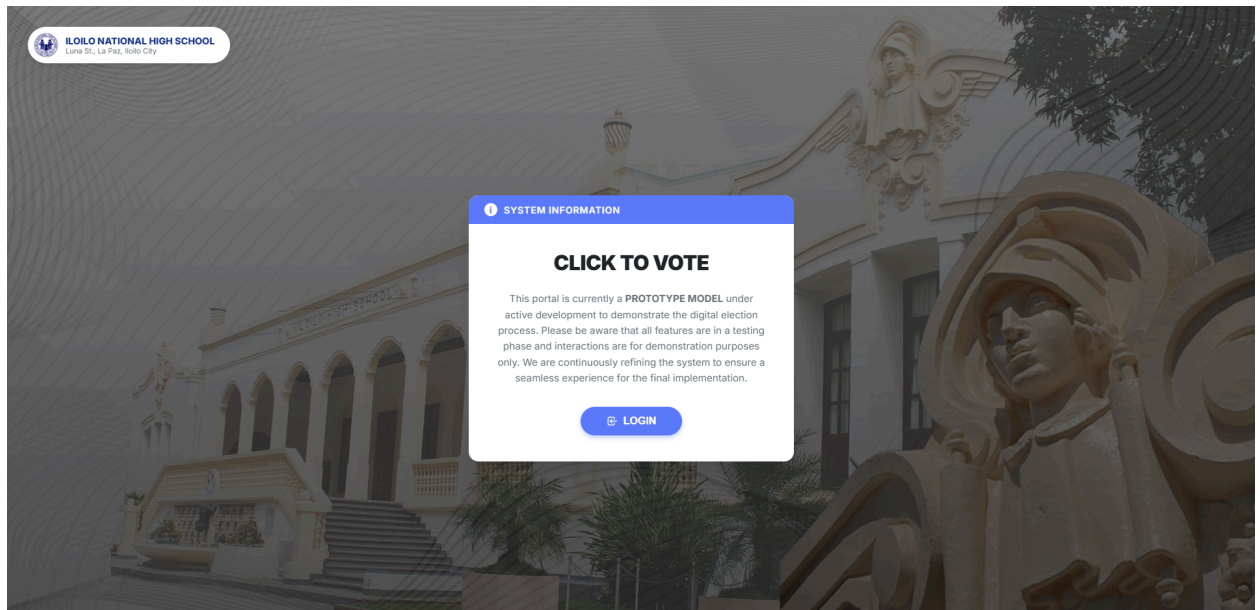
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Development

After the design of the system diagrams, proper development of Click to Vote actually started. With the front-end screens first being made which include: Login, Election Dashboard, Election History, Voter Management, Candidate Management, Officer Management and Audits. After the screens are all in place, back-end development started with Supabase then was integrated to be run from a local host using apache and mySQL from Xampp.

Figure 10

Login Screen



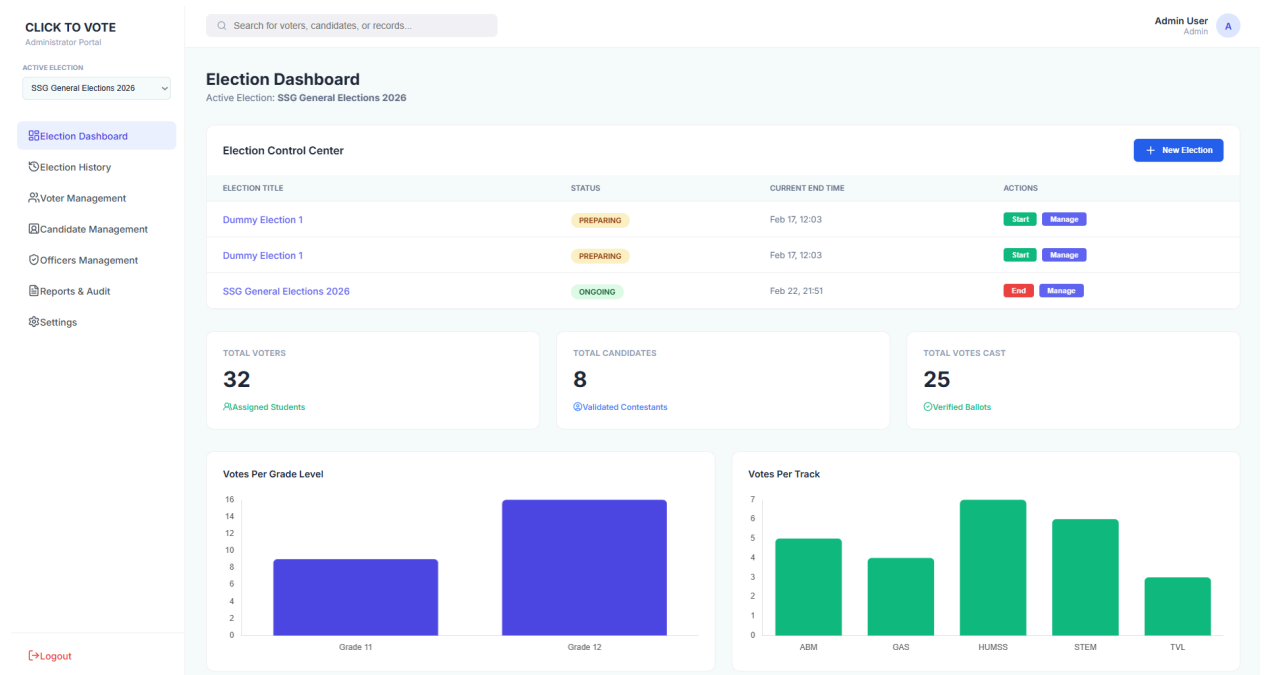


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Figure 11

Election Dashboard





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Figure 12

Election History

CLICK TO VOTE
Administrator Portal

ACTIVE ELECTION
SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

[>]Logout

Search for voters, candidates, or records...

Admin User
Admin

Admin User

Election History

Voter turnout and candidate results per election year

Jump to School Year

Dummy Election 1

GENERATE RESULTS

PREPARING

Dummy Election 1

GENERATE RESULTS

PREPARING

SSG General Elections 2026

GENERATE RESULTS

ONGOING



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Figure 13

Voter Management

CLICK TO VOTE
Administrator Portal

ACTIVE ELECTION
SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

Logout

Search for voters, candidates, or records...

Admin User
Admin

Voters List
Managing voters for SSG General Elections 2026

TOTAL VOTERS
32

VOTERS WHO VOTED
25

VOTERS WHO HAVEN'T VOTED
7

+ ADD VOTER

Import CSV

SEARCH
Search by email

TRACK
All Tracks

GRADE
All Grades

USER ID	NAME	EMAIL	TRACK	GRADE	SECTION	STATUS	ACTIONS
26-0009	Student 9	student9@ilolionhs.edu.ph	STEM	Grade 11	C	Voted	
26-0010	Student 10	student10@ilolionhs.edu.ph	HUMSS	Grade 11	B	Voted	
26-0011	Student 11	student11@ilolionhs.edu.ph	TVL	Grade 11	D	Pending	
26-0012	Student 12	student12@ilolionhs.edu.ph	GAS	Grade 11	B	Voted	
26-0013	Student 13	student13@ilolionhs.edu.ph	ABM	Grade 11	A	Voted	
26-0014	Student 14	student14@ilolionhs.edu.ph	HUMSS	Grade 11	D	Voted	
26-0015	Student 15	student15@ilolionhs.edu.ph	TVL	Grade 11	C	Pending	



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Figure 14

Candidate Management

CLICK TO VOTE
Administrator Portal

ACTIVE ELECTION
SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

Logout

Search for voters, candidates, or records...

Admin User
Admin

ONGOING

Candidate Management
Managing SSG General Elections 2026

+ ADD CANDIDATE

DEFINE POSITION

DEFINE PARTY

TOTAL CANDIDATES
8

UNIQUE POSITIONS
4

ACTIVE PARTIES
0

Candidates by Position

President 2
Vice President 2
Secretary 2
Treasurer 2

Candidates by Party
No parties defined.

SEARCH
Search candidates...

POSITION
All Positions

PARTY
All Parties

CANDIDATE	POSITION	PARTY	GRADE/TRACK	ACTIONS
Student 1 26-0001 student1@iloilohns.edu.ph	PRESIDENT	Alpha Party	Grade 12 STEM	
Student 2 26-0002 student2@iloilohns.edu.ph	PRESIDENT	Beta Party	Grade 12 HUMSS	
Student 3 26-0003 student3@iloilohns.edu.ph	VICE PRESIDENT	Alpha Party	Grade 12 GAS	



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Figure 15

Officer Management

CLICK TO VOTE
Administrator Portal

ACTIVE ELECTION
SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

Logout

Search for voters, candidates, or records...

Admin User
Admin

Officer Management
Personnel for SSG General Elections 2026

Assign New Officer to Election

FULL NAME
Enter name

STUDENT ID / PERSONNEL ID
XX-XXXX

EMAIL
email@school.edu

ROLE
Officer

PASSWORD
ilolohns

ASSIGN TO ELECTION

Admins
0 ACTIVE

No admins assigned.

Advisers
0 ACTIVE

No advisers assigned.

COMEAs Officers
0 ACTIVE

No officers assigned.

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Figure 16

Audits

CLICK TO VOTE
Administrator Portal

ACTIVE ELECTION
SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

Logout

Search for voters, candidates, or records...

Admin User
Admin

Reports & Audit Trail

Monitoring activity for SSG General Elections 2026

TIMESTAMP	USER	ACTION	DETAILS
Feb 16, 2026 04:07:09	Admin User Admin 100-00003	ADDED POSITION	No additional details
Feb 16, 2026 04:05:13	Admin User Admin 100-00003	ELECTION_UPDATED	Updated election 3078b7d1-8f4f-4cb9-8129-71c0ed99c28e
Feb 16, 2026 04:03:45	Admin User Admin 100-00003	CREATED ELECTION	No additional details
Feb 16, 2026 04:03:13	Admin User Admin 100-00003	CREATED ELECTION	No additional details
Feb 16, 2026 03:40:11	Admin User Admin 100-00003	REGISTERED VOTER	No additional details
Feb 16, 2026 03:39:30	Admin User Admin 100-00003	VOTER_REMOVED	Removed voter ID 168178a2-bf8b-4a0d-b7ed-bd1abaccb664 from election 536319a7-f28c-476e-9348-e66dcd1dd2a6e
Feb 16, 2026 03:07:56	Jay Orly Mil Santiago Voter 123-00023	CAST BALLOT	No additional details
Feb 15, 2026 23:54:43	Admin User Admin 100-00003	VOTER_UPDATED	Updated voter ID 168178a2-bf8b-4a0d-b7ed-bd1abaccb664 info
Feb 15, 2026 22:01:25	Admin User Admin 100-00003	REGISTERED VOTER	No additional details
Feb 15, 2026 21:59:30	Admin User Admin 100-00003	VOTER_REMOVED	Removed voter ID e85babf2-56b0-4712-888a-eba98f2d9496 from election 536319a7-f28c-476e-9348-e66dcd1dd2a6e



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Evaluation

After the development of Click to Vote, it was then presented to 2 sets of respondents, 1 group of Non-IT Experts and another group of IT Experts. These groups will then evaluate the system based on 2 international standardized tools used to evaluate the quality and usability of a system. ISO/IEC Software Quality and Requirements Evaluation (SQuaRE) Model and the System Usability Scale (SUS).

Respondents of the Study

As stated in chapter 1, the respondents of this study will consist of INHS-SHS Elections Committee Officers. These Officers, which will be referred to in the study as “Non-IT Experts,” were selected using the convenience sampling method, additionally, selected INHS faculty from the Information Technology (IT) department or any similar department, referred to in the study as “IT Experts,” have chosen by the purposive sampling method will also be respondents of this study. Both “Non-IT Experts” and “IT Experts” will become evaluators of the system, chosen because of their direct involvement in an election or how the software behind such a system would function.

Table 2

Table of Distribution of Respondents

	N	%
Total	13	100%
IT Experts	6	42.86%
Non-IT Experts	7	57.14%



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Instrumentation

To achieve the objectives of the study, the researchers utilized questionnaires as the primary tool for data collection. The questionnaires, in Likert scales, were designed to gather feedback and evaluations from the respondent. Two research questionnaires were used in this study:

Table 3

Likert Scale for the SQuaRE Model.

SCALE	VERBAL INTERPRETATION
4.21 - 5.00	Excellent
3.41 - 4.20	Very Good
2.61 - 3.40	Good
1.81 - 2.60	Poor
1.00 - 1.80	Very Poor

ISO/IEC 25010:2011 Software Quality Evaluation Checklist. This instrument was used to evaluate the system's quality in terms of functionality, usability, reliability, and performance efficiency. This uses a 5-point Likert scale.



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Table 4

Likert Scale for the SUS.

Category	Value
Strongly Agree	5
Agree	4
Neutral	3
Disagree	2
Strongly Disagree	1

System Usability Scale (SUS). The SUS questionnaire was used to measure the perceived usability of Click to Vote based on the SHS Election Committee's feedback. It consists of statements rated using a 5-point Likert scale along with its rating results; its corresponding values are shown below.



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Table 5
Rating System for the SUS.

Grade	SUS Score	Percentile Range	Adjective	Acceptability
A+	84.1 - 100	96 - 100	Best Imagineable	Acceptable
A	80.8 - 84	90 - 95	Excellent	Acceptable
A-	78.9 - 80.7	85 - 89	Excellent	Acceptable
B+	77.2 - 78.8	80 - 84	Excellent	Acceptable
B	74.1 - 77.1	70 - 79	Excellent	Acceptable
B-	72.6 - 74	65 - 69	Excellent	Acceptable
C+	71.1 - 72.5	60 - 64	Good	Acceptable
C	65 - 71	41 - 59	Good	Marginal
C-	62.7 - 64.9	35 - 40	Good	Marginal
D	51.7 - 62.6	15 - 34	Ok	Marginal
F	25.1 - 51.6	2 - 14	Poor	Not Acceptable
F	0 - 25	0 - 1.9	Worst Imagineable	Not Acceptable

The SUS uses a specific rating system for determining the "usability" of any system using the "SUS Score" which is determined by the ratings the respondents give in the questionnaire after some calculations which will be discussed in data analysis. This table presents the legend that will determine how Click to Vote's usability will be evaluated.



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Data Gathering Procedure

The data-gathering procedure of the study was conducted via researcher prepared google forms based on the ISO/IEC SQuaRE Model and the System Usability Scale while keeping in accordance with the descriptive-developmental research design and followed the ADDE model. This approach allowed the researchers to systematically develop the system while describing and evaluating its quality and usability.

Data Analysis Procedure

The data gathered from respondents were used to find means and the standard deviations before ultimately interpreting the results via a legend provided by each tool used were systematically analyzed to ensure accurate and meaningful results.. Responses were divided into those from IT experts and Non-IT experts.

For the ISO/EID SQuaRE Model Evaluation. The means for every category in the SQuaRE Model were divided into their own tables and computed to determine the overall evaluation of responses to the questionnaire, while the standard deviation was used to assess their variability.

For the System Usability Scale (SUS), the scoring method follows a structured calculation process. For odd-numbered items (1, 3, 5, 7, and 9), 1 is subtracted from the



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respondent's given rating. Conversely, for even-numbered items (2, 4, 6, 8, and 10), the rating is subtracted from 6, and the difference was subtracted from 5. After adjusting all responses, the scores are summed, and the total is then multiplied by 2.5 to obtain the final SUS score, which ranges from 0 to 100.

To methodize data management, the researchers utilized Microsoft Excel for encoding the evaluation ratings and organizing the dataset. Additionally, the researchers utilized a custom statistical system created using JavaScript.



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CHAPTER IV

RESULTS AND DISCUSSION

This chapter reviews the results and discussion of the quantitative data and findings of the study. Provides a comprehensive discussion of their significance within the research domain.

Table 6

Evaluation Results of the SQuaRE Model for IT Experts

Subcategory	N	Mean	SD	Interpretation
FUNCTIONALITY	6.00	4.72	0.68	Excellent
Completeness	6.00	4.67	0.82	Excellent
Correctness	6.00	4.67	0.82	Excellent
Appropriateness	6.00	4.83	0.41	Excellent
EFFICIENCY	6.00	4.72	0.68	Excellent
Time Behavior	6.00	4.67	0.82	Excellent
Resource Utilization	6.00	4.67	0.82	Excellent
Efficiency Compliance	6.00	4.83	0.41	Excellent
COMPATIBILITY	6.00	4.42	0.80	Excellent
Co-existence	6.00	4.50	0.55	Excellent
Interoperability	6.00	4.33	1.21	Excellent
USABILITY	6.00	4.75	0.23	Excellent
Appropriateness recognizeability	6.00	4.50	0.84	Excellent
Learnability	6.00	4.83	0.41	Excellent
Operability	6.00	4.83	0.41	Excellent
User Error Protection	6.00	4.33	0.52	Excellent
User Interface Aesthetics	6.00	5.00	0.00	Excellent
Accessibility	6.00	5.00	0.00	Excellent



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Subcategory	N	Mean	SD	Interpretation
RELIABILITY	6.00	4.58	0.47	Excellent
Maturity	6.00	4.83	0.41	Excellent
Availability	6.00	4.83	0.41	Excellent
Fault Tolerance	6.00	4.33	0.52	Excellent
Recoverability	6.00	4.33	0.82	Excellent
SECURITY	6.00	4.43	0.69	Excellent
Confidentiality	6.00	4.50	0.84	Excellent
Integrity	6.00	4.50	0.84	Excellent
Non-repudiation	6.00	4.33	0.82	Excellent
Accountability	6.00	4.33	0.82	Excellent
Authenticity	6.00	4.50	0.55	Excellent
MAINTAINABILITY	6.00	4.57	0.43	Excellent
Modularity	6.00	4.67	0.52	Excellent
Reusability	6.00	4.83	0.41	Excellent
Analyzability	6.00	4.50	0.55	Excellent
Modifiability	6.00	4.17	1.17	Very Good
Testability	6.00	4.67	0.82	Excellent
PORTABILITY	6.00	4.78	0.40	Excellent
Adaptability	6.00	4.67	0.52	Excellent
Installability	6.00	4.83	0.41	Excellent
Replaceability	6.00	4.83	0.41	Excellent
TOTAL	6.00	4.62	0.49	Excellent

Note: Very Poor - 1.00 - 1.80, Poor - 1.81 - 2.60, Good - 2.61 - 3.40, Very Good - 3.41 - 4.20, Excellent - 4.21 - 5.00



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On the Functional Stability

The evaluation for functional stability primarily focuses on whether the system developed covered all specified tasks and objectives as well as provides correct results. The mean for all questions equaled 4.90 (SD = 0.21) which was interpreted as “Excellent.”

On the Performance Efficiency

The evaluation for performance efficiency focuses on the efficiency to which the system uses its available resources to produce results in a timely and non-resource intensive manner. The mean for performance efficiency returned as 4.74 (SD = 0.47) which was interpreted as “Excellent.”

On the Compatibility

The evaluation for compatibility focuses on how well the system works with other already existing systems, what can be done with both systems working together and if Click to Vote is beneficial or detrimental to a previously existing system. The mean for compatibility returned as 4.62 (SD = 0.62) which was interpreted as “Excellent.”



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On the Usability

The evaluation for compatibility focuses on how well the system interfaces and interacts with its users, making its purpose clear and allowing for users to easily use the system while preventing errors from occurring. The mean for usability returned as 4.81 (SD = 0.23) which was interpreted as “Excellent.”

On the Reliability

The evaluation for reliability focuses on how well the system functions under unfavorable conditions as well as its ability to recover data after an interruption. The mean for reliability returned as 4.50 (SD = 0.43) which was interpreted as “Excellent.”

On the Security

The evaluation for security focuses on how well the system protects its data and only reveals such data to authorized users, as well as logging when such a user accesses that data for transparency. The mean for security returned as 4.57 (SD = 0.53) which was interpreted as “Excellent.”



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On the Maintainability

The evaluation for maintainability focuses on how well the system's components can be modified without negatively impacting the performance or functionality of the system as a whole. The mean for maintainability returned as 4.63 (SD = 0.36) which was interpreted as "Excellent."

On the Portability

The evaluation for portability focuses on how well the system can easily be installed or transferred to different hardware or software as well as possibly new environments altogether. The mean for portability returned as 4.74 (SD = 0.39) which was interpreted as "Excellent."

Overall System Quality

Overall, when getting the mean score of all the categories in the SQuaRE model evaluation, Click to Vote received an overall score of 4.69 which is interpreted as "Excellent." This means that the evaluation of Click to Vote, determined by 13 participants including IT and Non-IT Experts, is of "Excellent" quality. This means Click to Vote was well thought out and an effective system for its specific tasks.



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Table 7:

Evaluation Results of the SUS for both IT and Non-IT Experts

User	N	SUS Score	Interpretation
Non-IT Expert 1	7	100.00	Acceptable
Non-IT Expert 2	7	90.00	Acceptable
Non-IT Expert 3	7	95.00	Acceptable
Non-IT Expert 4	7	82.50	Acceptable
Non-IT Expert 5	7	95.00	Acceptable
Non-IT Expert 6	7	80.00	Acceptable
Non-IT Expert 7	7	90.00	Acceptable
IT Expert 1	6	97.50	Acceptable
IT Expert 2	6	77.50	Acceptable
IT Expert 3	6	100.00	Acceptable
IT Expert 4	6	100.00	Acceptable
IT Expert 5	6	95.00	Acceptable
IT Expert 6	6	100.00	Acceptable
Total		92.50	Acceptable

Legend: Not Acceptable = 0 - 51.6; Marginal = 51.7 - 71; Acceptable = 71.1 - 100

The evaluation results the System Usability Scale for both IT and Non-IT Experts returned a mean of 92.50 which was interpreted as “Acceptable,” this indicates Click to Vote’s ease of use/user friendliness in its design. Interestingly however, the lowest score provided was from IT Expert 2 who provided a mean score of 77.50.



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CHAPTER V

SUMMARY, CONCLUSION, IMPLICATIONS AND RECOMMENDATIONS

The section comprising summary, conclusions, implications, and recommendations provides a comprehensive overview of the key findings and insights derived from the study. Its primary goal is to offer a concise yet informative summary of the research, emphasizing the main outcomes. Additionally, this section explores the practical implications of the findings, providing valuable insights into their real-world applications and significance.

Summary

The design and development of Click to Vote would provide a secure and efficient alternative to traditional and Google Form elections. Click to Vote underwent extensive analysis including feedback and multiple interviews with the clientele, rigorous design and development processes and evaluation of both system quality using the ISO/IEC SQuaRE Model and system usability using the SUS which returned as “Excellent” and “Acceptable” respectively.



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Results and Findings

Click to Vote was distributed and tested by 7 SHS Election Committee Officers and 6 IT experts who were shown the functionalities of the system. It was evaluated according to overall software quality and system usability. Within overall software quality via ISO/IEC, Click to Vote was scored as “Excellent”. Which can be further broken down into categories:

For functionality, Click to Vote was described as “Excellent” (mean = 4.90). For performance efficiency, Click to Vote was described as “Excellent” (mean = 4.74). For compatibility, Click to Vote was described as “Excellent” (mean = 4.62). For usability, Click to Vote was described as “Excellent” (mean = 4.81). For reliability, Click to Vote was described as “Excellent” (mean = 4.50). For security, Click to Vote was described as “Excellent” (mean = 4.57). For maintainability, Click to Vote was described as “Excellent” (mean = 4.63); and for portability, Click to Vote was described as “Excellent” (mean = 4.74).

For system usability, the SUS was used. All of the evaluators have provided scores of 77.5 or higher which was described as “Acceptable.” This showcases the level of detail that went into Click to Vote’s overall design, allowing for a wide array of people with



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varying technical skills to still be able to use the system, proving it to be intuitive and an effective system if it were to be deployed.

Conclusion

From the results of the 2 evaluations that Click to Vote underwent, ISO/IEC has determined that Click to Vote is of "excellent" quality with thorough thought and extensive consideration put into its analysis, design, and development. As for the SUS, results indicated that Click to Vote is an intuitive system to use and is easily understandable by all of its users.

To conclude this study, the researchers have managed to accomplish the following:

1. Developed a web-based voting system for the SSLG-SHS election.
2. Evaluated the overall quality of the developed system based on the ISO/IEC 25010:2011 Systems and Software Quality Requirements and Evaluation (SQuaRE) Model.
3. Evaluated the usability of the developed system using the System Usability Scale (SUS).



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Implications

Theoretically, from the developed system as well as the evaluation that preceded it, Click to Vote can be used as a baseline for future systems to follow. Given the evaluation returned that Click to Vote as being of “Excellent” quality and “Acceptable” usability, Click to Vote has laid the groundwork for a potential future election system not just for the SHSLG Election in INHS, but also possibly to elections as a whole in many different schools, not just limited to INHS or the SHSLG.

Practically, Click to Vote has set a foundation for further systems to build off of. Before Click to Vote was created, there were no current systems about online elections for Senior High School. There have been many for colleges and national elections but never in high school. Click to Vote was the first system specifically designed for INHS SHS and its uses could be extended to reach the entirety of INHS and even other schools should it be developed further in the future.



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Recommendations

1. Expand the Scope of the System

Future studies may explore expanding the scope of Click to Vote beyond Senior High School Learner Government Elections. This could involve developing it into a more general voting system capable of supporting various types of democratic elections.

2. Incorporate Alternative Voting Methods

Another potential area for development is the inclusion of additional voting methods. Currently, Click to Vote uses the First Past the Post (Majority Rule) system, where each voter casts one vote per position. Future versions could incorporate alternative voting systems such as the Single Transferable Vote or the Alternative Vote.

3. Scale the System to a National Level

Lastly, future research could examine the possibility of expanding Click to Vote to a national scale. While school elections are important within their context, implementing online voting systems for national elections could significantly increase convenience and accessibility for voters, potentially transforming how democratic processes are conducted.



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References

- Bugayong, R., et al. (2025). *Click to Vote: Security and Integrity of Online Election in a Public Senior High School [Unpublished manuscript]*. Iloilo National High School.
- Chouhan, S., & Sharma, G. (2025). *A new era of elections: Leveraging blockchain for fair and transparent voting*. arXiv. <https://arxiv.org/abs/2502.16127>
- de los Santos, G. N., de los Santos, J. R. N., & de los Santos, L. G. (2020, November 1). *e-voting kiosk: A Network Architecture School-based Registration and Voting System*. IEEE Xplore. <https://doi.org/10.1109/HNICEM51456.2020.9400135>
- Debant, A., Hirschi, L. (2023). *The Security of Electronic Voting: Vulnerabilities and Solutions*
<https://www.inria.fr/en/security-electronic-voting-digital-security-confidentiality>
- Department of Education (2023, May 5) *Interim Guidelines of the Learners Government Programs (LGP) for School Year 2023-2024 [Memorandum]*. Department of Education
https://www.deped.gov.ph/wp-content/uploads/FO-2023-04-003_Interim-Guidelines-of-the-Learner-Government-Program-LGP-for-School-Year-2023-2024.pdf
- Fantozzi, P., Iecher, M., Laura, L., Naldi, M., & Ruggetti, V. (2025). *Electronic voting worldwide: The state of the art*. *Information*, 16(8), 650.
<https://doi.org/10.3390/info16080650>
- ID.ee. (2025, October 21). *E-voting and e-elections*
<https://www.id.ee/en/article/e-voting-and-e-elections>
- Kurt, S. (2018, December 16). *ADDIE model: Instructional design*. *Educational Technology*.
<https://educationaltechnology.net/the-addie-model-instructional-design>
- LibGuides (n.d.). *Design and Development Research (DDR) for instructional design*.
<https://resources.nu.edu/c.php?g=1013605&p=8395580>



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- Norris, P. (2014). *Why electoral integrity matters*. Cambridge University Press.
- Official Gazette. (1987). Article 3, 1987 Philippine Constitution. *Official Gazette*.
<http://www.officialgazette.gov.ph/constitutions/1987-constitution/>
- Pugeda, R. (2025). *E-VOTING SYSTEM: ENHANCING STUDENT ELECTIONS AND GOVERNANCE THROUGH DIGITALIZATION – The GLOW*. The-Glow.ph.
<https://www.the-glow.ph/e-voting-system-enhancing-student-elections-and-governance-through-digitalization>
- Republic Act No. 7166 (Philippines). (1991). An Act Providing for Synchronized National and Local Elections and for Electoral Reforms. Official Gazette.
<https://www.officialgazette.gov.ph/>
- Republic Act No. 9369 (Philippines). (2007). An Act to Further Modernize the Election Process, Repealing for the Purpose Batas Pambansa Blg. 881 and for Other Purposes. Official Gazette. <https://www.officialgazette.gov.ph/>
- Tabanao, E. G., & Cece, J. C. (2025). *Development of student government e-voting application*. Journal of Interdisciplinary Perspectives, 3(2).
<https://doi.org/10.69569/jip.2024.0675>
- Tahir, N. A. N., Jamil, F., & Abd Rahman, N. H. (2025). *Enhancing student election integrity through a secure online voting system*. National EngiTech Digest, 2(1).
<https://jktss.puo.edu.my/jurnalpuo/index.php/NED/article/view/249>



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APPENDIX A

Communication



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December 10, 2025

DELORAH CECILIA L. FANTILLO
Principal IV
Office of the Principal

Thru: **YVES KIM C. CABANTING**
ICT Department Head

Greetings of peace!

The undersigned are Grade 12 Technical-Vocational-Livelihood-Information and Communications Technology (TVL-ICT) learners undertaking our subjects in Inquiries, Investigations and Immersion (III) as well as Computer Programming. We are conducting a developmental study titled **“Click to Vote: A Digital Voting System for a Public Senior High School.”** The said study aims to develop a system to automate elections within INHS to aid in the efficiency and accuracy of an election.

In line with this, we respectfully ask for your permission to request Senior High School (SHS) learner's information for the school year 2025-2026. This data will be used as security measures to verify the identity of all learners and prevent outsiders from logging in. We assure you that all data collected will be treated with the utmost confidentiality and handled in accordance with the Data Privacy Act of 2012.

We sincerely look forward to your favorable response. Thank you very much for your kind support and consideration. Should you need additional information, we can be contacted via our leader's contact number 0919 706 1648.

Respectfully yours,

JAY ORLY MIL L. SANTIAGO
Research Group Leader

REZZO V. BUGAYONG
Researcher

ZUELYKA V. CASTRO
Researcher

XIAN MICHAEL D. PALOMAR
Researcher

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Noted:

BERNADETTE R. ALBISO
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Research Adviser, TVL-ICT

Recommending Approval:

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Assistant School Principal II for Academics
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Thru: **JUN JUN SENON**
SHS COMEA Adviser

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In line with this, we respectfully ask for your permission to conduct a formal interview with our school's Senior High School Committee on Elections and Appointments (SHS COMEA) Adviser. This interview aims to gather information regarding the process on how elections are conducted as well as any current system already implemented in INHS.

We sincerely look forward to your favorable response. Thank you very much for your kind support and consideration. Should you need additional information, we can be contacted via our leader's contact number 0919 706 1648.

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JO-AN R. PET
TVL/TechPro Track Coordinator

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TVL/TechPro Java Programming NCIII Teacher

Greetings of peace!

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In line with this, the researchers respectfully ask for your permission to be one of the **IT Experts** who will evaluate the system based on the **Software Quality Requirements and Evaluation (SQuaRe) Model and the System Usability Scale (SUS)** through a survey provided by the researchers. Data from this research will be kept tight and secured. The help you may extend to the researchers will greatly contribute to the completion of the study.

We sincerely look forward to your favorable response. Thank you very much for your kind support and consideration. Should you need additional information, we can be contacted via our leader's contact number 0919 706 1648.

Respectfully yours,

JAY ORLY MIL L. SANTIAGO
Research Group Leader

REZZO V. BUGAYONG
Researcher

ZUELYKA V. CASTRO
Researcher

XIAN MICHAEL D. PALOMAR
Researcher

KURT LEOVINCE G. TSE WING
Researcher

CLARK RHAZEN G. YMBANG
Researcher

Noted:

BERNADETTE R. ALBISO
Research Adviser, TVL-ICT

AL JOHN D. ARTACHO
Research Adviser, TVL-ICT

Approved:

GLEN DAVE S. HORBINO
Assistant School Principal II for Academics
Senior High School



SENIOR HIGH SCHOOL



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January 11, 2026

JOSEPH R. GAREZA
ICT Department

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AL JOHN D. ARTACHO
Research Adviser, TVL-ICT

Approved:

GLEN DAVE S. HORBINO
Assistant School Principal II for Academics
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ROEL S. PALMAIRA
ICT Department

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AL JOHN D. ARTACHO
Research Adviser, TVL-ICT

Approved:

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Assistant School Principal II for Academics
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Dear Respondents

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Your participation is **completely voluntary**. **All responses will remain confidential**. Data from this research will be kept tight and secured. The help you may extend to the researchers will greatly contribute to the completion of the study.

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AL JOHN D. ARTACHO
Research Adviser, TVL-ICT

Approved:

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Assistant School Principal II for Academics
Senior High School



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APPENDIX B

Title Approval Sheet



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INVESTIGATION, INQUIRIES, IMMERSION
TITLE APPROVAL SHEET

SECTION: TVL-ICT 12A

DATE: December 8, 2025

GROUP MEMBERS:

REZZO V. BUGAYONG
ZUELYKA V. CASTRO
XIAN MICHAEL D. PALOMAR

JAY ORLY MIL L. SANTIAGO
KURT LEOVINCE G. TSE WING
CLARK RHAZEN G. YMBANG

Proposed title:

Click to Vote: A Digital Voting System for a Public Senior High School

☐ Approved

☐ Not Approved

BERNADETTE R. ALBISO
Name & Signature

AL JOHN D. ARTACHO
Name & Signature

JIMMY D. CABRIERTO
Name & Signature

ROEL S. PALMAIRA
Name & Signature

LEO C. ABE ABE
Name & Signature



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APPENDIX C

Research Instrument



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January 11, 2026

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CLARK RHAZEN G. YMBANG
Researcher

Noted:

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Research Adviser, TVL-ICT

AL JOHN D. ARTACHO
Research Adviser, TVL-ICT

Approved:

GLEN DAVE S. HORBINO
Assistant School Principal II for Academics
Senior High School



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Part 1. RESPONDENT'S PERSONAL INFORMATION

Name (Optional) : _____ Date Evaluated: _____

Position: ☐ IT Expert Specialization: _____

Part 2. QUESTIONNAIRE PROPER FOR IT EXPERT

The questions below are adopted from **ISO/IEC 25010** Software Quality Model Characteristics.

Instruction:

Please carefully evaluate and rate each item and mark check (✓) in the box corresponding to your response using the following numerical equivalents.

Rating Scale

5	=	Excellent
4	=	Very Good
3	=	Good
2	=	Fair
1	=	Poor

Characteristics and Sub characteristics					
<i>Functional Characteristics</i>	1	2	3	4	5
Completeness The degree to which the set of functions covers all the specified tasks and user objectives					
Correctness The degree to which a product or system provides the correct results with the needed degree of precision					
Appropriateness The degree to which the functions facilitate the accomplishment of specified tasks and objectives					
<i>Efficiency Characteristics</i>	1	2	3	4	5
Time behavior The degree to which the response and processing times and throughput rates of a product or system, when performing its functions, meet requirements					



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Resource utilization The degree to which the amounts and types of resources a product or system uses, when performing its functions, meet requirements.					
Efficiency compliance The capability of the software to provide the required performance, relative to the amount of resources used, under stated conditions					
Compatibility Characteristics	1	2	3	4	5
Co-existence The degree to which a product can perform its required functions efficiently while sharing a common environment and resources with other products without detrimental impact on any other product					
Interoperability The degree to which two or more systems, products, or components can exchange information and use the information that has been exchanged					
Usability Characteristics	1	2	3	4	5
Appropriateness recognizability The degree to which users can recognize whether a product or system is appropriate for their needs					
Learnability The degree to which specified users can use a product or system to achieve specified learning goals to use the product or system with effectiveness, efficiency, freedom from risk, and satisfaction in a specified context.					
Operability The degree to which a product or system has attributes that make it easy to operate and control					
User error protection The degree to which a system protects users against making errors					
User interface aesthetics The degree to which a user interface enables pleasing and satisfying interaction for the user					
Accessibility The degree to which a product or system can be used by people with the broadest range of characteristics and capabilities to achieve a specified goal in a specified context of use					



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<i>Reliability Characteristics</i>	1	2	3	4	5
Maturity The degree to which a system, product, or component meets needs for reliability under regular operation					
Availability The degree to which a system, product, or component is operational and accessible when required for use					
Fault tolerance The degree to which a system, product, or component operates as intended despite the presence of hardware or software faults					
Recoverability The degree to which, in the event of an interruption or a failure, a product or system can recover the data directly affected and re-establish the desired state of the system					
<i>Security Characteristics</i>	1	2	3	4	5
Confidentiality The degree to which a product or system ensures that data are accessible only to those authorized to have access					
Integrity The degree to which a system, product, or component prevents unauthorized access to, or modification of, computer programs or data					
Non-repudiation The degree to which actions or events can be proven to have taken place so that the events or actions cannot be repudiated later					
Accountability The degree to which the actions of an entity can be traced uniquely to the entity					
Authenticity The degree to which the identity of a subject or resource can be proved to be the one claimed					
<i>Maintainability Characteristics</i>	1	2	3	4	5
Modularity The degree to which a system or computer program is composed of discrete components such that a change to one component has minimal impact on other components					



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Reusability The degree to which an asset can be used in more than one system or in building other assets					
Analyzability The degree of effectiveness and efficiency with which it is possible to assess the impact on a product or system of an intended change to one or more of its parts, or to diagnose a product for deficiencies or causes of failures, or to identify parts to be modified					
Modifiability The degree to which a product or system can be effectively and efficiently modified without introducing defects or degrading existing product quality					
Testability The degree of effectiveness and efficiency with which test criteria can be established for a system, product, or component, and tests can be performed to determine whether those criteria have been met.					
Portability Characteristics	1	2	3	4	5
Adaptability The degree to which a product or system can effectively and efficiently be adapted for different or evolving hardware, software, or other operational or usage environments					
Installability The degree of effectiveness and efficiency with which a product or system can be successfully installed and uninstalled in a specified environment					
Replaceability The degree to which a product can replace another specified software product for the same purpose in the same environment					
TOTAL					



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QUESTIONNAIRE FOR NON-IT PARTICIPANTS

Part 1. RESPONDENTS PERSONAL INFORMATION

Name (Optional): _____ Date Evaluated: _____

Position: _____

Part 2. QUESTIONNAIRE PROPER

Instruction: Please carefully evaluate and rate each item and mark check (✓) in the box corresponding to your response using the following numerical equivalents.

Rating Scale

- | | | |
|---|---|-----------|
| 5 | = | Excellent |
| 4 | = | Very Good |
| 3 | = | Good |
| 2 | = | Fair |
| 1 | = | Poor |

Criteria	1	2	3	4	5
1. I would like to use this system frequently					
2. The system is easy to understand and not overly complex.					
3. The system was easy to use.					
4. I can use the system without needing technical support.					
5. The various functions of this system are well-integrated					
6. The system consistently performs as expected, without issues.					
7. I would imagine that most people would learn to use this system very quickly					
8. The system is efficient to use.					
9. I feel confident using the system.					
10. Overall, I am satisfied with this system					
TOTAL					

Signature

Thank you, and God bless!



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APPENDIX D

Raw Data



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Evaluation Results on ISO/EIC SQuaRE Model for IT Experts

Subcategory	Sample	IT 1	IT 2	IT 3	IT 4	IT 5	IT 6	Mean	SD	Interpretation
FUNCTIONALITY	6.00	5.00	3.33	5.00	5.00	5.00	5.00	4.72	0.68	Excellent
Completeness	6.00	5.00	3.00	5.00	5.00	5.00	5.00	4.67	0.82	Excellent
Correctness	6.00	5.00	3.00	5.00	5.00	5.00	5.00	4.67	0.82	Excellent
Appropriateness	6.00	5.00	4.00	5.00	5.00	5.00	5.00	4.83	0.41	Excellent
EFFICIENCY	6.00	5.00	3.33	5.00	5.00	5.00	5.00	4.72	0.68	Excellent
Time Behavior	6.00	5.00	3.00	5.00	5.00	5.00	5.00	4.67	0.82	Excellent
Resource Utilization	6.00	5.00	3.00	5.00	5.00	5.00	5.00	4.67	0.82	Excellent
Efficiency Compliance	6.00	5.00	4.00	5.00	5.00	5.00	5.00	4.83	0.41	Excellent
COMPATIBILITY	6.00	5.00	3.00	4.50	5.00	4.00	5.00	4.42	0.80	Excellent
Co-existence	6.00	5.00	4.00	4.00	5.00	4.00	5.00	4.50	0.55	Excellent
Interoperability	6.00	5.00	2.00	5.00	5.00	4.00	5.00	4.33	1.21	Excellent
USEABILITY	6.00	4.67	4.33	4.83	5.00	4.83	4.83	4.75	0.23	Excellent
Appropriateness recognizeability	6.00	4.00	3.00	5.00	5.00	5.00	5.00	4.50	0.84	Excellent
Learnability	6.00	5.00	4.00	5.00	5.00	5.00	5.00	4.83	0.41	Excellent
Operability	6.00	5.00	5.00	5.00	5.00	5.00	4.00	4.83	0.41	Excellent
User Error Protection	6.00	4.00	4.00	4.00	5.00	4.00	5.00	4.33	0.52	Excellent
User Interface Aesthetics	6.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	0.00	Excellent
Accessibility	6.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	0.00	Excellent
RELIABILITY	6.00	5.00	3.75	4.50	4.75	4.50	5.00	4.58	0.47	Excellent
Maturity	6.00	5.00	4.00	5.00	5.00	5.00	5.00	4.83	0.41	Excellent
Availability	6.00	5.00	4.00	5.00	5.00	5.00	5.00	4.83	0.41	Excellent
Fault Tolerance	6.00	5.00	4.00	4.00	4.00	4.00	5.00	4.33	0.52	Excellent
Recoverability	6.00	5.00	3.00	4.00	5.00	4.00	5.00	4.33	0.82	Excellent
SECURITY	6.00	5.00	3.20	4.80	4.40	4.20	5.00	4.43	0.69	Excellent
Confidentiality	6.00	5.00	3.00	5.00	5.00	4.00	5.00	4.50	0.84	Excellent
Integrity	6.00	5.00	3.00	5.00	5.00	4.00	5.00	4.50	0.84	Excellent
Non-repudiation	6.00	5.00	3.00	4.00	4.00	5.00	5.00	4.33	0.82	Excellent
Accountability	6.00	5.00	3.00	5.00	4.00	4.00	5.00	4.33	0.82	Excellent
Authenticity	6.00	5.00	4.00	5.00	4.00	4.00	5.00	4.50	0.55	Excellent
MAINTAINABILITY	6.00	5.00	3.80	4.40	4.60	4.80	4.80	4.57	0.43	Excellent
Modularity	6.00	5.00	5.00	4.00	5.00	5.00	4.00	4.67	0.52	Excellent
Reusability	6.00	5.00	5.00	4.00	5.00	5.00	5.00	4.83	0.41	Excellent
Analyzability	6.00	5.00	4.00	4.00	4.00	5.00	5.00	4.50	0.55	Excellent
Modifiability	6.00	5.00	2.00	5.00	4.00	4.00	5.00	4.17	1.17	Very Good
Testability	6.00	5.00	3.00	5.00	5.00	5.00	5.00	4.67	0.82	Excellent
PORTABILITY	6.00	5.00	4.67	5.00	5.00	4.00	5.00	4.78	0.40	Excellent
Adaptability	6.00	5.00	4.00	5.00	5.00	4.00	5.00	4.67	0.52	Excellent
Installability	6.00	5.00	5.00	5.00	5.00	4.00	5.00	4.83	0.41	Excellent
Replaceability	6.00	5.00	5.00	5.00	5.00	4.00	5.00	4.83	0.41	Excellent
TOTAL	6.00	4.96	3.68	4.75	4.84	4.54	4.95	4.62	0.49	Excellent

Evaluation Result on System Usability

User No.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Sum of Even	Sum of Odd	SUS Score	Remark
Non-IT Expert 1	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	20.00	20.00	100.00	Acceptable
Non-IT Expert 2	4.00	5.00	5.00	4.00	5.00	3.00	5.00	5.00	5.00	5.00	17.00	19.00	90.00	Acceptable
Non-IT Expert 3	5.00	5.00	5.00	5.00	5.00	4.00	5.00	5.00	4.00	5.00	19.00	19.00	95.00	Acceptable
Non-IT Expert 4	5.00	5.00	4.00	4.00	4.00	4.00	3.00	5.00	4.00	5.00	18.00	15.00	82.50	Acceptable
Non-IT Expert 5	4.00	5.00	5.00	5.00	5.00	4.00	5.00	5.00	5.00	5.00	19.00	19.00	95.00	Acceptable
Non-IT Expert 6	5.00	4.00	4.00	3.00	5.00	4.00	4.00	4.00	4.00	5.00	15.00	17.00	80.00	Acceptable
Non-IT Expert 7	4.00	4.00	5.00	5.00	4.00	5.00	5.00	5.00	4.00	5.00	19.00	17.00	90.00	Acceptable
IT Expert 1	5.00	5.00	5.00	4.00	5.00	5.00	5.00	5.00	5.00	5.00	19.00	20.00	97.50	Acceptable
IT Expert 2	3.00	5.00	5.00	5.00	2.00	5.00	4.00	4.00	5.00	3.00	17.00	14.00	77.50	Acceptable
IT Expert 3	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	20.00	20.00	100.00	Acceptable
IT Expert 4	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	20.00	20.00	100.00	Acceptable
IT Expert 5	4.00	5.00	5.00	5.00	5.00	4.00	5.00	5.00	5.00	5.00	19.00	19.00	95.00	Acceptable
IT Expert 6	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	20.00	20.00	100.00	Acceptable
Total	4.54	4.85	4.85	4.62	4.62	4.46	4.69	4.85	4.69	4.85	18.62	18.38	92.50	Acceptable



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APPENDIX E

Gantt Chart



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ID	Task	Start Date	Due Date	Duration	Nov	December							January			February	
					W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	
3	Development	01/13/2026	02/06/2026	3 Weeks													
3.1	Coding of Frontend Screens	01/13/2026	02/1/2026	3 Weeks													
3.2	Connection of Front-end Screens	01/28/2026	01/31/2026	4 Days													
3.4	Database Schema Creation	01/29/2026	01/31/2026	3 Days													
3.3	Database to Frontend Connection	02/04/2026	02/06/2026	3 Days													
3.5	Presentation to COMEA Adviser	02/06/2026	02/06/2026	1 Day													
3.6	Consultation with IT Expert again	02/06/2026	02/06/2026	1 Day													
3.7	Creation of Online Evaluation Tools	02/03/2026	02/03/2026	1 Day													
4	Evaluation	02/06/2026	02/10/2026	5 Days													
4.1	Presentation of System to Non-IT and IT Professionals	02/06/2026	02/06/2026	1 Day													
4.2	Evaluation Proper	02/06/2026	02/06/2026	1 Day													
4.3	Summarization of Evaluation Results	02/06/2026	02/07/2026	2 Days													
4.4	Recording of Results to Manuscript	02/07/2026	02/07/2026	1 Day													
4.5	Summarization and Conclusion of Results	02/07/2026	02/07/2026	1 Day													
4.6	Final Presentation Preparation	02/08/2026	02/10/2026	3 Days													
4.7	Final Presentation	02/10/2026	02/10/2026	1 Day													
5	Documentation	11/27/2025	02/10/2026	3 Months													



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ID	Task	Start Date	Due Date	Duration	Nov	December			January				February				
					W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13
1	Analysis	11/27/2025	12/12/2025	3 Weeks													
1.1	Identification of Research Problem	11/27/2025	11/27/2025	1 Day													
1.2	Review of Studies Related to the Problem	11/27/2025	12/10/2025	3 Weeks													
1.3	Interview with COMEA Adviser	11/29/2025	12/03/2025	2 Week													
1.4	Researching of Related Systems	11/29/2025	12/05/2025	2 Week													
1.5	Consultation with an IT expert	12/02/2025	12/02/2025	1 Day													
1.7	Prototype Creation	12/02/2025	12/03/2025	2 Days													
1.8	Title Presentation	12/06/2025	12/06/2025	1 Day													
1.9	Writing of Letters	12/10/2025	12/10/2025	1 Day													
1.10	Creation of Diagrams (Pseudocode, Flowchart, Context Flow, Data Flow, Use Case)	12/10/2025	12/12/2025	3 Days													
2	Design	12/02/2025	02/06/2026														
2.1	Writing of Methodology	12/15/2025	02/06/2026	10 Weeks													
2.2	Designing of System Flow	12/02/2025	12/15/2026	2 Weeks													
2.3	Designing of GUI	12/15/2026	02/01/2026	9 Weeks													



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APPENDIX F

Flowchart of Click to Vote

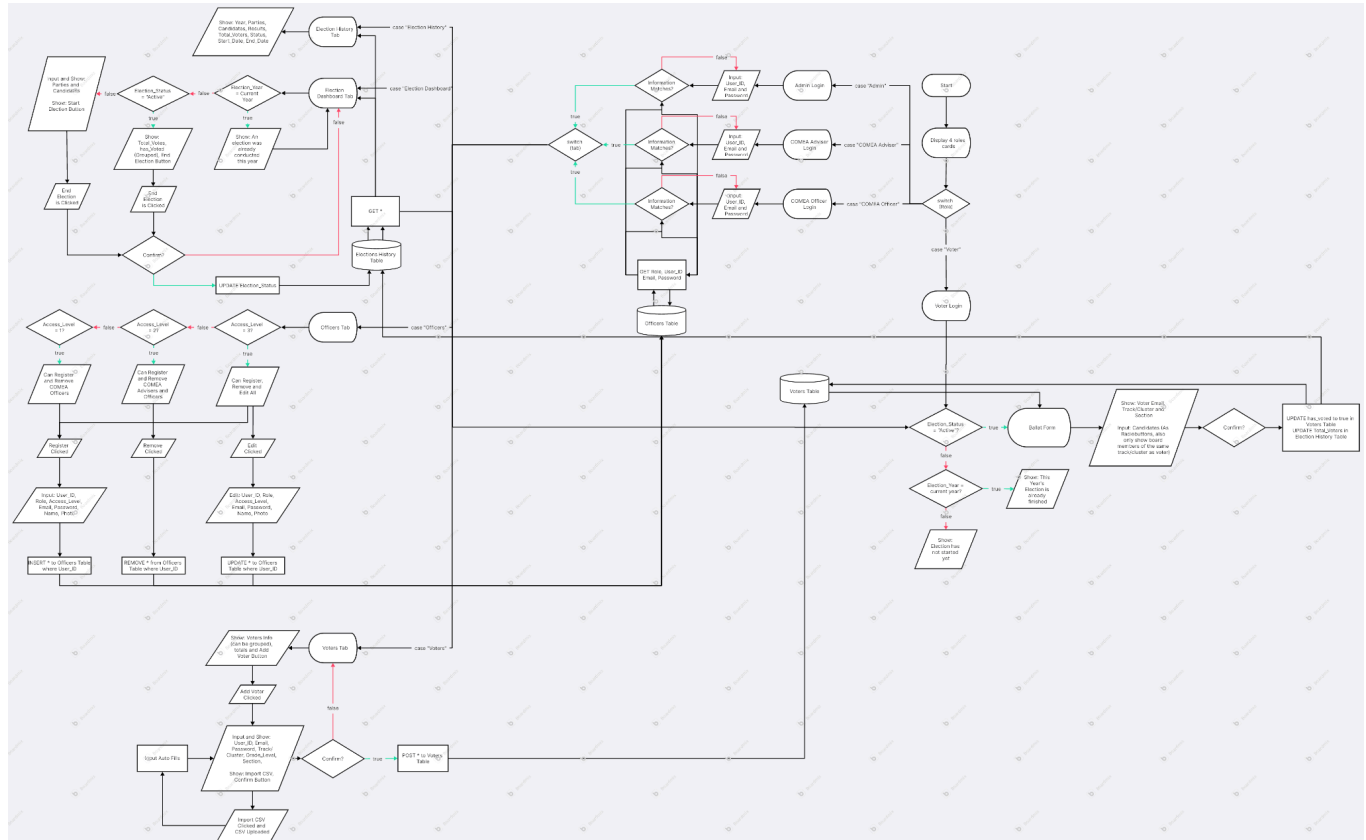


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APPENDIX G

Context Flow Diagram

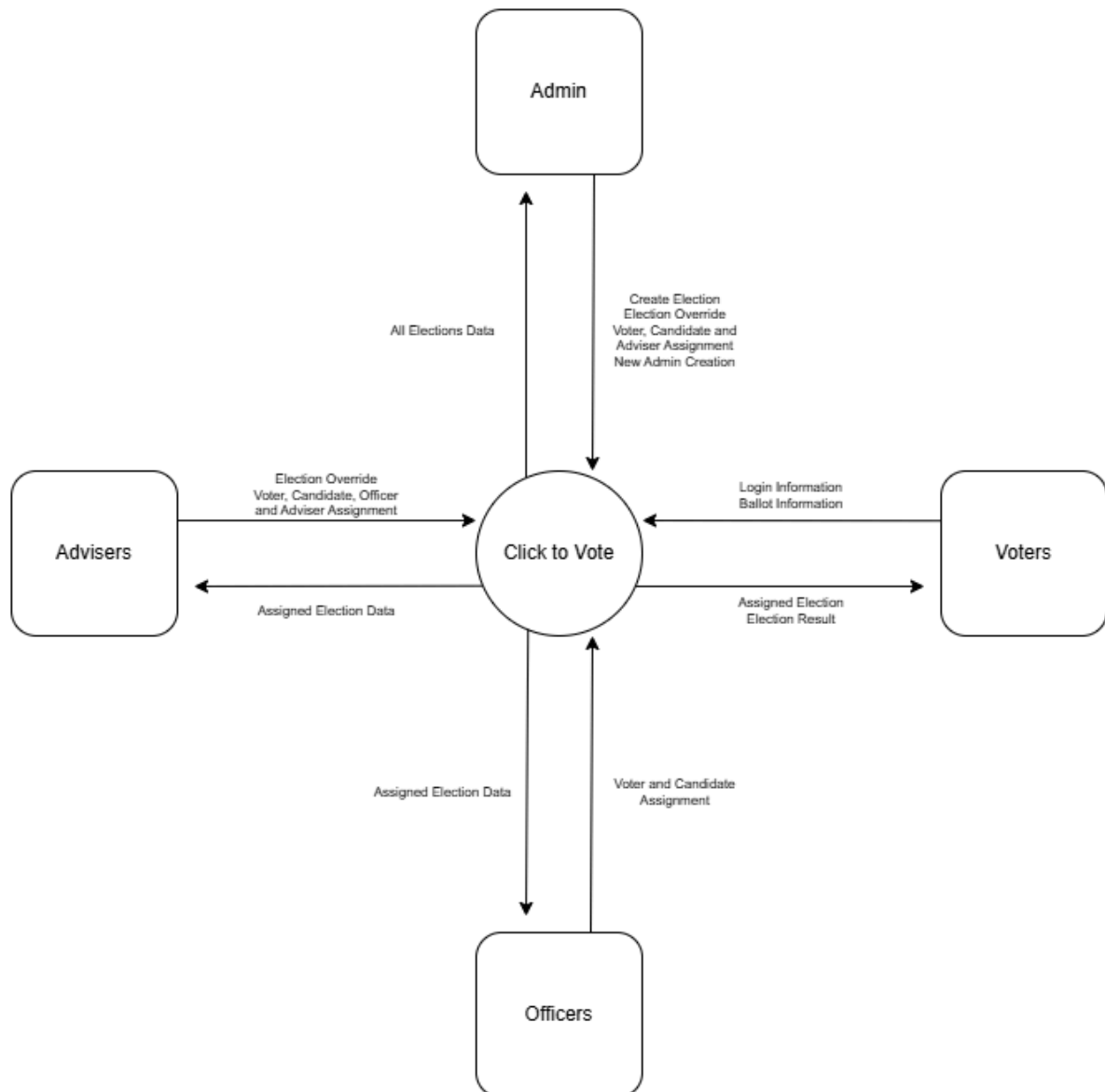


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APPENDIX H

Data Flow Diagram

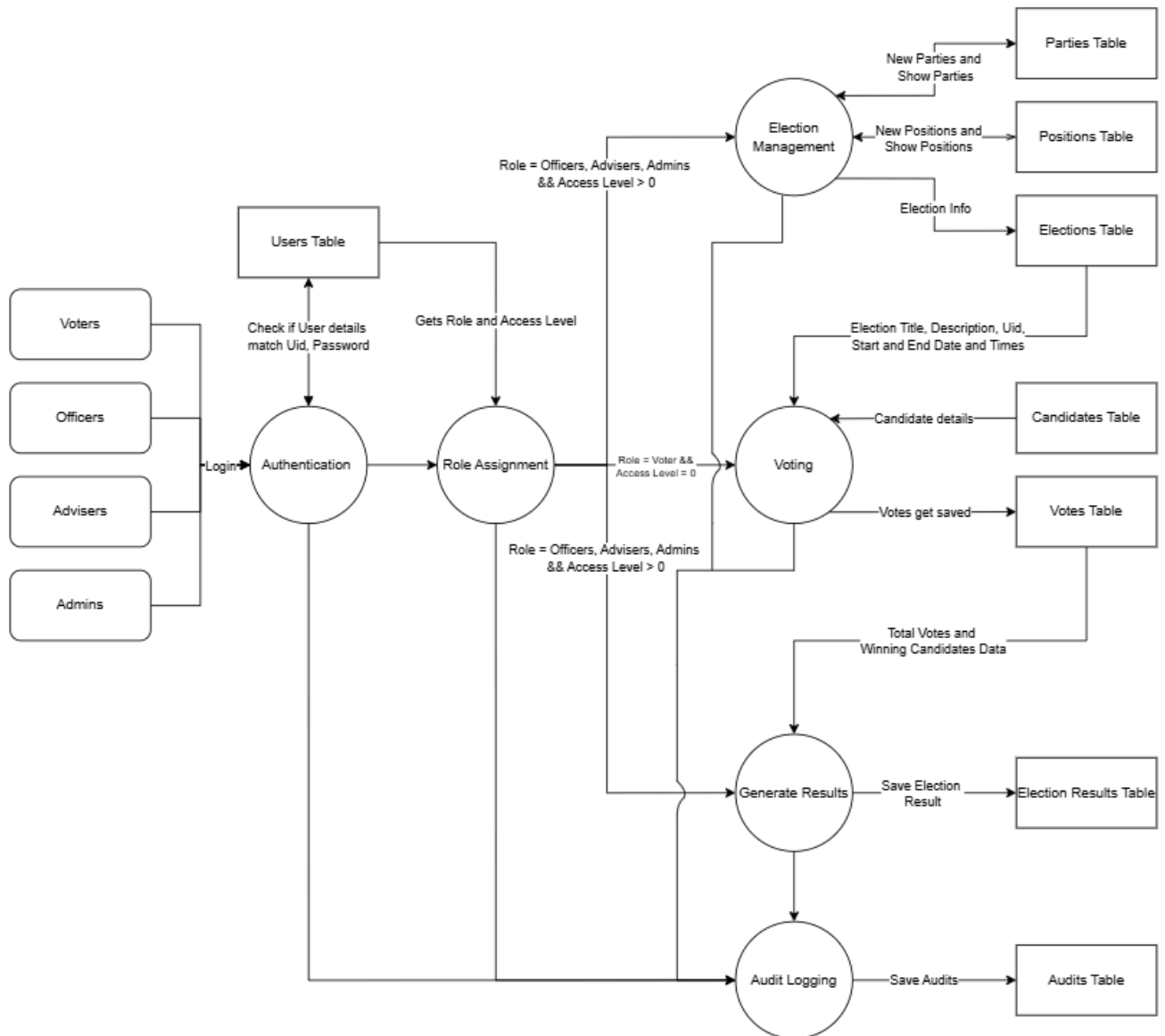


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APPENDIX I

Use Case Diagram



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APPENDIX J

Entity Relationship Diagram

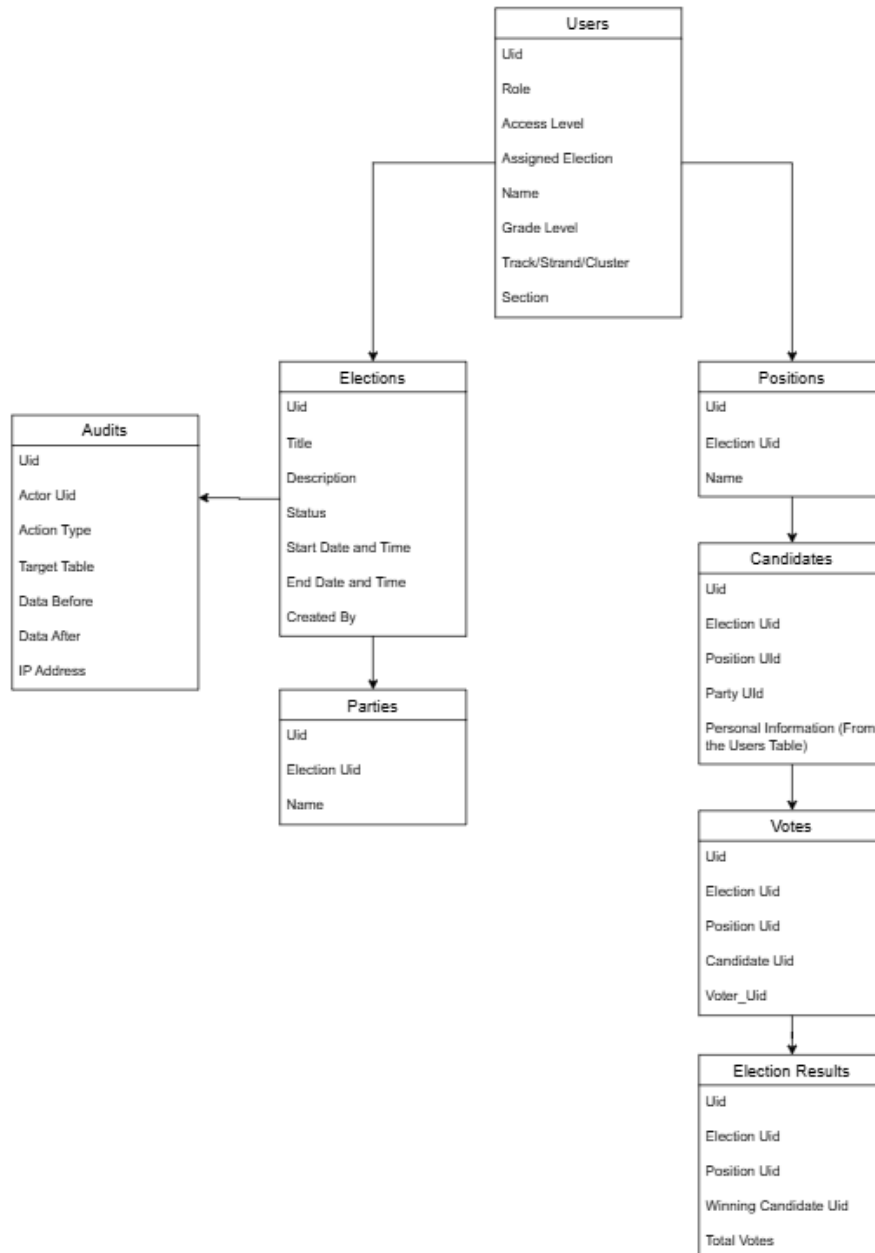


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APPENDIX K

System Screenshots - Login Screen

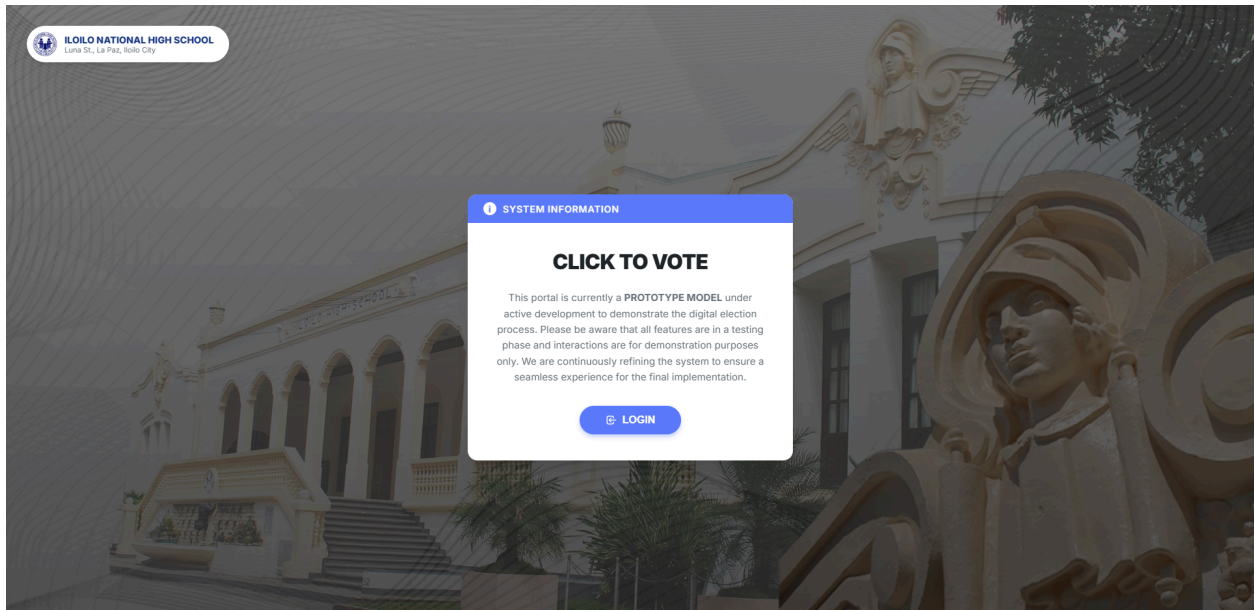


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APPENDIX L

System Screenshots - Election Dashboard



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CLICK TO VOTE
Administrator Portal

ACTIVE ELECTION

SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

Logout

Search for voters, candidates, or records...

Election Dashboard

Active Election: SSG General Elections 2026

Election Control Center

+ New Election

ELECTION TITLE	STATUS	CURRENT END TIME	ACTIONS
Dummy Election 1	PREPARING	Feb 17, 12:03	Start Manage
Dummy Election 1	PREPARING	Feb 17, 12:03	Start Manage
SSG General Elections 2026	ONGOING	Feb 22, 21:51	End Manage

TOTAL VOTERS

32

Assigned Students

TOTAL CANDIDATES

8

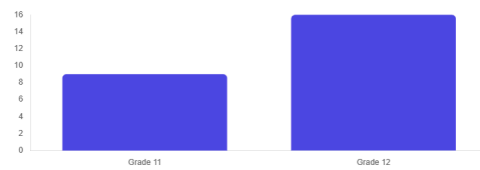
Validated Contestants

TOTAL VOTES CAST

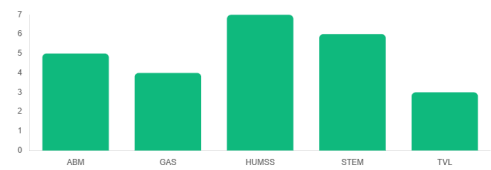
25

Verified Ballots

Votes Per Grade Level



Votes Per Track



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APPENDIX M

System Screenshots - Election History



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CLICK TO VOTE
Administrator Portal

ACTIVE ELECTION

SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

Logout

Search for voters, candidates, or records...

Admin User
Admin

Voters List

Managing voters for SSG General Elections 2026

TOTAL VOTERS

32

VOTERS WHO VOTED

25

VOTERS WHO HAVEN'T VOTED

7

+ ADD VOTER

Import CSV

SEARCH			TRACK		GRADE		
Search by email			All Tracks		All Grades		
USER ID	NAME	EMAIL	TRACK	GRADE	SECTION	STATUS	ACTIONS
26-0009	Student 9	student9@ilolionhs.edu.ph	STEM	Grade 11	C	Voted	Edit Delete
26-0010	Student 10	student10@ilolionhs.edu.ph	HUMSS	Grade 11	B	Voted	Edit Delete
26-0011	Student 11	student11@ilolionhs.edu.ph	TVL	Grade 11	D	Pending	Edit Delete
26-0012	Student 12	student12@ilolionhs.edu.ph	GAS	Grade 11	B	Voted	Edit Delete
26-0013	Student 13	student13@ilolionhs.edu.ph	ABM	Grade 11	A	Voted	Edit Delete
26-0014	Student 14	student14@ilolionhs.edu.ph	HUMSS	Grade 11	D	Voted	Edit Delete
26-0015	Student 15	student15@ilolionhs.edu.ph	TVL	Grade 11	C	Pending	Edit Delete



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APPENDIX N

System Screenshots - Voter Management



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Administrator Portal

ACTIVE ELECTION

SSG General Elections 2026

Election Dashboard

Election History

Voter Management

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Reports & Audit

Settings

Logout

Search for voters, candidates, or records...

Admin User
Admin

Voters List

Managing voters for SSG General Elections 2026

TOTAL VOTERS

32

VOTERS WHO VOTED

25

VOTERS WHO HAVEN'T VOTED

7

+ ADD VOTER

Import CSV

SEARCH			TRACK		GRADE		
Search by email			All Tracks		All Grades		
USER ID	NAME	EMAIL	TRACK	GRADE	SECTION	STATUS	ACTIONS
26-0009	Student 9	student9@iloiionhs.edu.ph	STEM	Grade 11	C	Voted	Edit Delete
26-0010	Student 10	student10@iloiionhs.edu.ph	HUMSS	Grade 11	B	Voted	Edit Delete
26-0011	Student 11	student11@iloiionhs.edu.ph	TVL	Grade 11	D	Pending	Edit Delete
26-0012	Student 12	student12@iloiionhs.edu.ph	GAS	Grade 11	B	Voted	Edit Delete
26-0013	Student 13	student13@iloiionhs.edu.ph	ABM	Grade 11	A	Voted	Edit Delete
26-0014	Student 14	student14@iloiionhs.edu.ph	HUMSS	Grade 11	D	Voted	Edit Delete
26-0015	Student 15	student15@iloiionhs.edu.ph	TVL	Grade 11	C	Pending	Edit Delete



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APPENDIX O

System Screenshots - Candidate Management



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CLICK TO VOTE
Administrator Portal

ACTIVE ELECTION
SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

Search for voters, candidates, or records...

Admin User
Admin

ONGOING

Candidate Management
Managing SSG General Elections 2026

+ ADD CANDIDATE

DEFINE POSITION

DEFINE PARTY

TOTAL CANDIDATES
8

UNIQUE POSITIONS
4

ACTIVE PARTIES
0

Candidates by Position

Candidates by Party

SEARCH
Search candidates...

POSITION
All Positions

PARTY
All Parties

CANDIDATE	POSITION	PARTY	GRADE/TRACK	ACTIONS
Student 1 26-0001 student1@iloilohns.edu.ph	PRESIDENT	Alpha Party	Grade 12 STEM	
Student 2 26-0002 student2@iloilohns.edu.ph	PRESIDENT	Beta Party	Grade 12 HUMSS	
Student 3 26-0003 student3@iloilohns.edu.ph	VICE PRESIDENT	Alpha Party	Grade 12 GAS	

Logout



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APPENDIX P

System Screenshots - Officer Management



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CLICK TO VOTE

Administrator Portal

ACTIVE ELECTION

SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

Search for voters, candidates, or records...

Admin UserAdmin

Officer Management

Personnel for SSG General Elections 2026

Assign New Officer to Election

FULL NAME

STUDENT ID / PERSONNEL ID

EMAIL

ROLE

PASSWORD

Enter name

XX-XXXX

email@school.edu

Officer

ilolohns

ASSIGN TO ELECTION

Admins

0 ACTIVE

No admins assigned.

Advisers

0 ACTIVE

No advisers assigned.

COMEAs Officers

0 ACTIVE

No officers assigned.

Logout



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APPENDIX Q

System Screenshots - Audits



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CLICK TO VOTE
Administrator Portal

ACTIVE ELECTION

SSG General Elections 2026

Election Dashboard

Election History

Voter Management

Candidate Management

Officers Management

Reports & Audit

Settings

Logout

Search for voters, candidates, or records...

Admin User
Admin

Reports & Audit Trail

Monitoring activity for SSG General Elections 2026

TIMESTAMP	USER	ACTION	DETAILS
Feb 16, 2026 04:07:09	Admin User Admin (00-0000)	ADDED POSITION	No additional details
Feb 16, 2026 04:05:13	Admin User Admin (00-0000)	ELECTION_UPDATED	Updated election 3078b7d1-8f4f-4cb9-8129-71c0ed99c28e
Feb 16, 2026 04:03:45	Admin User Admin (00-0000)	CREATED ELECTION	No additional details
Feb 16, 2026 04:03:13	Admin User Admin (00-0000)	CREATED ELECTION	No additional details
Feb 16, 2026 03:40:11	Admin User Admin (00-0000)	REGISTERED VOTER	No additional details
Feb 16, 2026 03:39:30	Admin User Admin (00-0000)	VOTER_REMOVED	Removed voter ID 168178a2-bf8b-4a0d-b7ed-bd1abaccb664 from election 536319a7-f28c-476e-9348-e66dc1dd2a6e
Feb 16, 2026 03:07:56	Jay Orly Mii Santiago Voter (123-00123)	CAST BALLOT	No additional details
Feb 15, 2026 23:54:43	Admin User Admin (00-0000)	VOTER_UPDATED	Updated voter ID 168178a2-bf8b-4a0d-b7ed-bd1abaccb664 info
Feb 15, 2026 22:01:25	Admin User Admin (00-0000)	REGISTERED VOTER	No additional details
Feb 15, 2026 21:59:30	Admin User Admin (00-0000)	VOTER_REMOVED	Removed voter ID e85babf2-56b0-4712-888a-eba98f2d9496 from election 536319a7-f28c-476e-9348-e66dc1dd2a6e



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APPENDIX R

Source Code



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```
<?php                                     // Find ongoing elections that this

require_once 'auth_helper.php';           voter is assigned to

$user = get_user();                       $stmt = $pdo->prepare("

                                           SELECT e.* FROM elections e

                                           JOIN election_assignments ea ON

if (!$user) {                             e.id = ea.election_id

    include 'landing.php';                 WHERE ea.user_id = ?

    exit;                                 AND e.status = 'Ongoing'

}                                         AND e.archived = FALSE

if (in_array($user['role'], ['Admin',     AND e.end_date_and_time >

'Adviser', 'Officer'])) {               CURRENT_TIMESTAMP

    include 'dashboard.php';              ");

    exit;                                $stmt->execute([$user['id']]);

}                                         $activeElections = $stmt->fetchAll();

                                           // Filter out elections where the user

                                           has already voted

// Voter redirection logic               $votedElectionsStmt =

if ($user['role'] === 'Voter') {         $pdo->prepare("SELECT election_id
```



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```

FROM votes WHERE voter_id = ?");          exit;

                                          }

$votedElectionsStmt->execute([$user['id

]);                                          // If no eligible elections but they were

    $votedIds =                             assigned to some active ones, they've

    $votedElectionsStmt->fetchAll(PDO::F    already voted

    ETCH_COLUMN);                          if (count($eligibleElections) === 0

                                          && count($activeElections) > 0) {

    $eligibleElections =                    if (!isset($_GET['success'])) &&

    array_filter($activeElections,         !isset($_GET['error'])) {

    function($e) use ($votedIds) {         header("Location:

    return !in_array($e['id'], $votedIds); index.php?error=already_voted");

    });                                    exit;

                                          }

    if (count($eligibleElections) === 1) { }

    $singleElection =

    reset($eligibleElections);              // For voters, only show their assigned

    header("Location: ballot.php?id=" .    elections in the list

    $singleElection['id']);                $selections = $activeElections;
  
```





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```

} else {
    $selections = $pdo->query("SELECT *
FROM elections WHERE archived =
FALSE ORDER BY created_at
DESC")->fetchAll();
}

$projectDescription =
$_SERVER['PROJECT_DESCRIPTION
'] ?? 'Online Election System for Senior
High School';

?>

<!doctype html>

<html lang="en">

<head>

    <meta charset="utf-8" />

    <meta name="viewport"
content="width=device-width,
initial-scale=1" />

    <title>Election Dashboard | <?=
htmlspecialchars($projectDescription)
?></title>

    <meta name="description"
content="<?=
htmlspecialchars($projectDescription)
?>" />

    <link rel="preconnect"
href="https://fonts.googleapis.com">

    <link rel="preconnect"
href="https://fonts.gstatic.com"
crossorigin>

    <link
href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600;700&d
isplay=swap" rel="stylesheet">

    <link rel="stylesheet"
href="assets/css/style.css?v=<?= time()
?>">

```



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```
</head>                                btn-outline">Logout</a>

<?php $isVotedModalActive =            </div>

(isset($_GET['success']) &&              </nav>

$_GET['success'] === 'voted') ||

(isset($_GET['error']) &&                <div class="container" <?=

$_GET['error'] === 'already_voted'); ?> $isVotedModalActive ? 'style="display:

<body class="<?= ($user['theme'] ??     none;" : " ?>>

'light') === 'dark' ? 'dark-theme' : " ?>"> <?php if (isset($_GET['success']) &&

    <nav class="navbar" <?=              $_GET['success'] !== 'voted'): ?>

$isVotedModalActive ? 'style="display:    <div style="background: #dcfce7;

none;" : " ?>>                          color: #166534; padding: 1rem;

    <a href="index.php"                  border-radius: var(--radius); border: 1px

class="brand">Click to Vote</a>         solid #bbf7d0; margin-bottom: 1.5rem;

    <div>                                font-size: 0.875rem;">

        <span style="margin-right: 1rem;    Action completed successfully.

color: var(--text-muted);"><?=          </div>

htmlspecialchars($user['name']) ?> (<?=  <?php endif; ?>

$user['role'] ?>)</span>                <div class="header-actions">

    <a href="logout.php" class="btn       <div>
```



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```
<h1 style="margin: 0; font-size: 1.5rem;">Elections</h1>

<p style="margin: 0; color: var(--text-muted);">Manage your school elections and voting sessions.</p>

</div>

<?php if (in_array($user['role'], ['Admin', 'Adviser', 'Officer'])): ?>
    <a href="create_election.php" class="btn btn-outline" style="margin-top: 1rem;">Setup Election</a>
</div>

<?php endif; ?>

</div>

<div class="card">
    <?php if (empty($elections)): ?>
        <div style="text-align: center; padding: 2rem;">
            <p style="color: var(--text-muted);">No elections found.
            Create your first election to get started.</p>
        </div>
    <?php else: ?>
        <table class="table">
            <thead>
                <tr>
                    <th>Title</th>
                    <th>Status</th>
                    <th>Period</th>
                    <th>Actions</th>
                </tr>
            </thead>
            <tbody>
                <tr>
                    <td>+ New Election</td>
                    <td></td>
                    <td></td>
                    <td></td>
                </tr>
            </tbody>
        </table>
    </div>
    <?php endif; ?>
</div>
```



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<pre> </tr> </thead> <tbody> <?php foreach (\$selections as \$selection): ?> <tr> <td> <div style="font-size: <div style="font-weight: 600;"><?= htmlspecialchars(\$selection['title']) ?></div> <div style="font-size: 0.75rem; color: var(--text-muted);"><?= htmlspecialchars(\$selection['description']) ?></div> </td> <td> <span class="badge badge-<?= strtolower(\$selection['status']) </pre>	<pre> ?>"> <?= htmlspecialchars(\$selection['status']) ?> </td> <div style="font-size: 0.875rem;"> <?= date('M d, H:i', strtotime(\$selection['start_date_and_time'])) ?> - <?= date('M d, H:i', strtotime(\$selection['end_date_and_time'])) ?> </div> </td> <td> <a href="view_election.php?id=<?= </pre>
--	---



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\$election['id'] ?>" class="btn	
btn-outline" style="padding: 0.25rem	<!-- Voted Modal -->
0.5rem; font-size: 0.75rem;">View	<div id="votedModal"
<?php if (\$election['status']	class="modal-overlay <?=
=== 'Ongoing'): ?>	(isset(\$_GET['success']) &&
<a href="ballot.php?id=<?=	\$_GET['success'] === 'voted')
\$election['id'] ?>" class="btn	(isset(\$_GET['error']) &&
btn-primary" style="padding: 0.25rem	\$_GET['error'] === 'already_voted') ?
0.5rem; font-size: 0.75rem; background:	'active' : " ?>">
#166534;">Vote	<div class="confirm-modal"
<?php endif; ?>	style="text-align: center;">
</td>	<div class="modal-icon"
</tr>	style="background: #f0fdf4; color:
<?php endforeach; ?>	#10b981; margin: 0 auto 24px;">
</tbody>	<svg style="width: 32px; height:
</table>	32px;" fill="none"
<?php endif; ?>	stroke="currentColor" viewBox="0 0 24
</div>	24"
</div>	xmlns="http://www.w3.org/2000/svg">



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```
<path stroke-linecap="round"
stroke-linejoin="round"
stroke-width="3" d="M5 13L4 19
7"></path>
</svg>
</div>
<h2>Already Voted</h2>
<p>You have already submitted
your vote for this election. Thank you
for participating!</p>
<div style="margin-top: 32px;">
  <a href="logout.php" class="btn
btn-primary" style="display: block;
width: 100%; text-decoration: none;
padding: 12px 0; background: #4f46e5;
border-radius: 12px; font-weight: 700;
color: white;">Back to Login</a>
</div>
</div>
```

```
</div>
<style>
/* Modal Styles */
.modal-overlay {
display: none;
position: fixed;
top: 0;
left: 0;
right: 0;
bottom: 0;
background: rgba(15, 23, 42, 0.6);
backdrop-filter: blur(4px);
z-index: 2000;
align-items: center;
justify-content: center;
padding: 20px;
}
```



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```
.modal-overlay.active {  
    to { transform: scale(1)  
    display: flex;  
    translateY(0); opacity: 1; }  
}  
  
.confirm-modal {  
    background: white;  
    border-radius: 20px;  
    max-width: 450px;  
    width: 100%;  
    padding: 32px;  
    box-shadow: 0 25px 50px -12px  
    rgba(0, 0, 0, 0.25);  
    animation: modalIn 0.3s  
    cubic-bezier(0.34, 1.56, 0.64, 1);  
}  
  
@keyframes modalIn {  
    from { transform: scale(0.9)  
    translateY(20px); opacity: 0; }  
  
.modal-icon {  
    width: 64px;  
    height: 64px;  
    background: #eef2ff;  
    color: #4f46e5;  
    border-radius: 16px;  
    display: flex;  
    align-items: center;  
    justify-content: center;  
    margin-bottom: 24px;  
}  
  
.confirm-modal h2 {  
    margin: 0 0 12px 0;  
    font-size: 1.5rem;
```



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```

font-weight: 800;                                </body>

color: #1e293b;                                </html>

}                                                <?php

                                                require_once 'auth_helper.php';

.confirm-modal p {

  margin: 0 0 12px 0;                            $error = "";

  color: #64748b;                                if ($_SERVER['REQUEST_METHOD']

  font-size: 1rem;                               === 'POST') {

  line-height: 1.6;                             $student_id = $_POST['student_id']

}                                                ?? ";

</style>                                         $email = $_POST['email'] ?? ";

                                                $role = $_POST['role'] ?? ";

<footer style="text-align: center; color:         $password = $_POST['password'] ??

var(--text-muted); padding: 2rem; <?=         ";

$isVotedModalActive ? 'display: none;' :

" ?>">                                         // 1. Find user locally to verify

  &copy; <?= date('Y') ?> Click to Vote         student_id, email and role match

I High School Online Election System           $stmt = db()->prepare("SELECT *

</footer>                                       FROM users WHERE student_id = ?

```



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```
AND email = ? AND role = ?");                                // Migrate to Supabase

$stmt->execute([$student_id, $email,                          $supabaseUser =
$role]);                                                       SupabaseAuth::createUser($email,
                                                                $password);
                                                                if (!$supabaseUser['error']) {
if ($user) {                                                  $auth =
                                                                // 2. Authenticate with Supabase
                                                                SupabaseAuth::signIn($email,
                                                                $password);
SupabaseAuth::signIn($email,                                  }
$password);                                                  }
                                                                }

                                                                if ($auth['error']) {
                                                                // Check if user exists locally
                                                                if (!$auth['error']) {
with this password but not in Supabase                        // Update supabase_uid if
yet                                                            missing
                                                                if (empty($user['supabase_uid']))
                                                                {
(!empty($user['password_hash']) &&                            {
password_verify($password,                                    $supabase_uid =
$user['password_hash'])) {                                    $auth['data']['user']['id'] ?? null;
```



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```
$supd =                                $error = 'Invalid Credentials. Please

db()->prepare("UPDATE users SET        check your UID, Email, and Role. ';

supabase_uid = ? WHERE id = ?");      }

$supd->execute([$supabase_uid,         if ($error && isset($_POST['role'])
$user['id']));                          &&

}                                       str_contains($_SERVER['HTTP_REFERER'] ?? '', 'index.php')) {

$_SESSION['user_id'] =                 // Only redirect back if we actually
$user['id'];                           came from landing page modal

$_SESSION['user_role'] =               header('Location: index.php?error='
$user['role'];                         . urlencode($error));

header('Location: index.php');         exit;

exit;                                  }

} else {                              }

$error = 'Authentication failed: '.   ?>

$auth['error'];                        <!DOCTYPE html>

}                                       <html lang="en">

} else {                              <head>
```



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```
<meta charset="UTF-8">
<meta name="viewport"
content="width=device-width,
initial-scale=1.0">
<title>Login - Iloilo National High
School</title>
<link rel="preconnect"
href="https://fonts.googleapis.com">
<link rel="preconnect"
href="https://fonts.gstatic.com"
crossorigin>
<link
href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600;700;80
0;900&display=swap" rel="stylesheet">
<link rel="stylesheet"
href="assets/css/landing.css?v=<?
time() ?>">
<style>

.login-page-container {
min-height: 100vh;
display: flex;
justify-content: center;
align-items: center;
z-index: 2;
position: relative;
}
</style>
</head>
<body class="landing-page"
style="background-image:
url('assets/images/background.jpg?v=<?
filetime('assets/images/background.jpg'
) ?>');">
<div class="login-page-container">
<div class="login-modal"
style="display: block; position: static;
```



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```
animation: none;">                                </div>

<div class="modal-header">                            <div class="modal-body">

    <div                                              <?php if ($error): ?>

class="modal-header-content">                        <div style="background:

    <div class="header-icon">                        #fee2e2; color: #b91c1c; padding:

        <svg width="24"                             0.75rem; border-radius: 10px;

height="24" viewBox="0 0 24 24"                      margin-bottom: 1.5rem; font-size:

fill="none" stroke="currentColor"                  0.85rem; border: 1px solid #fecaca;

stroke-width="2"                                    text-align: center;">

stroke-linecap="round"                              <?=

stroke-linejoin="round"><path d="M17               htmlspecialchars($error) ?>

21v-2a4 4 0 0 0 4-4H5a4 4 0 0 0 4

4v2"></path><circle cx="9" cy="7"

r="4"></circle><path d="M23 21v-2a4

4 0 0 0 3-3.87"></path><path d="M16

3.13a4 4 0 0 1 0 7.75"></path></svg>

    </div>                                          <label>User

Type</label>

<h2>Election Login</h2>                            <div

</div>                                              class="input-container">
```



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```

    <i>
        value="Admin">Admin</option>
    </i>
    <svg width="18"
        height="18" viewBox="0 0 24 24"
        fill="none" stroke="currentColor"
        stroke-width="2"
        stroke-linecap="round"
        stroke-linejoin="round"><path d="M20
        21v-2a4 4 0 0 0-4-4H8a4 4 0 0 0-4
        4v2"></path><circle cx="12" cy="7"
        r="4"></circle></svg>
    </i>
    <select name="role">
        <option
            value="Voter">Voter</option>
        <option
            value="Officer">Officer</option>
        <option
            value="Adviser">Adviser</option>
        <option
            value="Admin">Admin</option>
    </select>
    </div>
    </div>
    <div class="form-group">
        <label>UID</label>
        <div
            class="input-container">
                <i>
                    <svg width="18"
                        height="18" viewBox="0 0 24 24"
                        fill="none" stroke="currentColor"
                        stroke-width="2"
                        stroke-linecap="round"
                        stroke-linejoin="round"><rect x="3"
                        y="4" width="18" height="16"
                        rx="2"></rect><line x1="7" y1="8"
                        x2="17" y2="8"></line><line x1="7"
  
```



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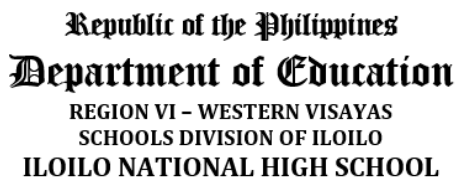
```

y1="12" x2="17" y2="12"></line><line
fill="none" stroke="currentColor"
x1="7" y1="16" x2="12"
stroke-width="2"
y2="16"></line></svg>
stroke-linecap="round"
</i>
stroke-linejoin="round"><path d="M4
4h16c1.1 0 2 .9 2 2v12c0 1.1-.9 2-2
2H4c-1.1 0-2-.9-2-2V6c0-1.1-.9-2-
2-2z"></path><polyline points="22,6
12,13 2,6"></polyline></svg>
</div>
</i>
</div>
<input type="email"
name="email"
placeholder="firstname.lastname@iloilo
nhs.edu.ph" required>
Account</label>
</div>
<div
class="input-container">
<i>
<div class="form-group">
<label>Password</label>
height="18" viewBox="0 0 24 24"
</div>

```



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```

class="input-container">
    <i>
        <svg width="18"
height="18" viewBox="0 0 24 24"
fill="none" stroke="currentColor"
stroke-width="2"
stroke-linecap="round"
stroke-linejoin="round"><rect x="3"
y="11" width="18" height="11" rx="2"
ry="2"></rect><path d="M7 11V7a5 5 0
0 1 10 0v4"></path></svg>
        </i>
    </div>
    <input
type="password" id="passwordInput"
name="password" placeholder="Enter
your password" required>
    <i>
        <svg width="18"
height="18" viewBox="0 0 24 24"
fill="none" stroke="currentColor"
stroke-width="2"

```



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```

stroke-linecap="round"                                </div>

stroke-linejoin="round"><path d="M15                </div>
3h4a2 2 0 0 1 2 2v14a2 2 0 0 1-2
2h-4"></path><polyline points="10 17                <script>
15 12 10 7"></polyline><line x1="15"                function togglePassword() {
y1="12" x2="3" y2="12"></line></svg>                const input =
    LOGIN                document.getElementById('passwordInp
    </button>                ut');
    </form>                const icon =
    <div class="text-center mt-3">                document.getElementById('eyeIcon');
    <small style="color:                if (input.type === 'password') {
var(--landing-text-muted); font-size:                input.type = 'text';
0.8rem;">Don't have an account? <a                icon.innerHTML = '<path
href="signup.php" style="color:                d="M17.94 17.94A10.07 10.07 0 0 1 12
var(--landing-primary); font-weight:                20c-7 0-11-8-11-8a18.45 18.45 0 0 1
600; text-decoration: none;">Register                5.06-5.94M9.9 4.24A9.12 9.12 0 0 1 12
here</a></small>                4c7 0 11 8 11 8a18.5 18.5 0 0 1-2.16
    </div>                3.19m-6.72-1.07a3 3 0 1
    </div>                1-4.24-4.24"></path><line x1="1"

```



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```

y1="1" x2="23" y2="23"></line>;
    } else {
        input.type = 'password';
        icon.innerHTML = '<path
d="M1 12s4-8 11-8 11 8 11 8-4 8-11
8-11-8-11-8z"></path><circle cx="12"
cy="12" r="3"></circle>;
    }
}
</script>
</body>
</html>
<?php
declare(strict_types=1);
require_once __DIR__ .
'/auth_helper.php';
require_login();
$user = get_user();

$id = $_GET['id'] ?? '';
if (!$id) {
    header("Location: index.php");
    exit;
}

$db = db();
$stmt = $pdo->prepare("SELECT *
FROM elections WHERE id = ?");
$stmt->execute([$id]);
$election = $stmt->fetch();

if (!$election) {
    die("Election not found.");
}

$positions = $pdo->prepare("SELECT *
FROM positions WHERE election_id =
? ORDER BY sort_order ASC");
  
```



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\$positions->execute([\$id]);	<link rel="preconnect"
\$positions = \$positions->fetchAll();	href="https://fonts.gstatic.com"
	crossorigin>
\$projectDescription =	<link
\$_SERVER['PROJECT_DESCRIPTION	href="https://fonts.googleapis.com/css2?
'] ?? 'Online Election System';	family=Inter:wght@400;500;600;700&d
?>	isplay=swap" rel="stylesheet">
<!doctype html>	<link rel="stylesheet"
<html lang="en">	href="assets/css/dashboard.css?v=<?= time() ?>">
<head>	<script
<meta charset="utf-8" />	src="https://unpkg.com/lucide@latest">
<meta name="viewport"	</script>
content="width=device-width,	</head>
initial-scale=1" />	<body class="dashboard-body <?= (\$user['theme'] ?? 'light') === 'dark' ? 'dark-theme' : " ?>">
<title><?= htmlspecialchars(\$election['title']) ?> Manage</title>	<?php require_once 'includes/sidebar.php'; ?>
<link rel="preconnect"	
href="https://fonts.googleapis.com">	



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```

<div style="display: flex; gap:
12px;">
<div class="main-wrapper">
  <?php require_once
  <?php if (in_array($user['role'],
  'includes/header.php'; ?>
  ['Admin', 'Adviser', 'Officer'])): ?>
    <button class="btn-action"
    style="background: #ffffff; border: 1px
    solid #e2e8f0; color: #4b5563; padding:
    10px 20px; border-radius: 8px;
    font-weight: 600; cursor: pointer;
    display: flex; align-items: center; gap:
    8px;"
    <h1 style="margin: 0; font-size:
    1.75rem; color: #1e293b;"><? =
    htmlspecialchars($selection['title'])
    ?></h1>
    <p class="welcome-msg"
    style="margin-top: 4px;"><? =
    htmlspecialchars($selection['description']
    ) ?></p>
  </div>
  <i data-lucide="edit-3"
  style="width: 16px;"></i>
  Edit Info
</button>
<?php if ($selection['status'] ===
'Preparing'): ?>

```



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```
<form
action="api/update_election_status.php"
method="POST">
    <input type="hidden"
name="id" value="<?= $election['id']
?>">
    <input type="hidden"
name="status" value="Ongoing">
    <button type="submit"
class="btn-action" style="background:
#166534; color: white; padding: 10px
20px; border: none; border-radius: 8px;
font-weight: 600; cursor: pointer;
display: flex; align-items: center; gap:
8px;">
        <i data-lucide="play"
style="width: 16px;"></i>
        Launch Election
    </button>
</form>
<?php elseif ($election['status']
=== 'Ongoing'): ?>
    <form
action="api/update_election_status.php"
method="POST">
        <input type="hidden"
name="id" value="<?= $election['id']
?>">
        <input type="hidden"
name="status" value="Finished">
        <button type="submit"
class="btn-action" style="background:
#ef4444; color: white; padding: 10px
20px; border: none; border-radius: 8px;
font-weight: 600; cursor: pointer;
display: flex; align-items: center; gap:
8px;">
            <i data-lucide="square"
```



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style="width: 16px;"></i>

<i

End Election

data-lucide="check-square"></i>

</button>

</div>

</form>

<span style="font-size: 0.75rem;

<?php endif; ?>

font-weight: 700; color: #64748b;

<?php endif; ?>

text-transform: uppercase; letter-spacing:

</div>

0.05em;">Status

</div>

</div>

<div class="stats-grid"

<div style="display: flex;

align-items: center; gap: 8px;">

style="grid-template-columns: repeat(3,

<span class="status-indicator

1fr); margin-bottom: 24px;">

<?= strtolower(\$selection['status']) ?>"

<div class="stat-card">

style="padding: 4px 12px;

<div style="display: flex;

border-radius: 9999px; font-size:

align-items: center; gap: 12px;

0.875rem; font-weight: 600;

margin-bottom: 16px;">

background: <?= \$selection['status'] ===

<div style="background: #eef2ff;

'Ongoing' ? '#dcfce7' : (\$selection['status']

padding: 10px; border-radius: 10px;

=== 'Preparing' ? '#fef9c3' : '#f1f5f9')

color: #4f46e5;">

?>; color: <?= \$selection['status'] ===



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'Ongoing' ? '#166534' : (\$election['status'] === 'Preparing' ? '#854d0e' : '#475569') ?>,"> <?= \$election['status'] ?> </div> </div> <div class="stat-card"> <div style="display: flex; align-items: center; gap: 12px; margin-bottom: 16px;"> <div style="background: #fff7ed; padding: 10px; border-radius: 10px; color: #f97316;"> <i data-lucide="calendar"></i> </div> Election Period </div> <div style="font-size: 0.875rem; font-weight: 600; color: #1e293b;"> <?= date('M d, H:i', strtotime(\$election['start_date_and_time'])) ?> — <?= date('M d, H:i', strtotime(\$election['end_date_and_time'])) ?> </div> </div> <div class="stat-card"> <div style="display: flex; align-items: center; gap: 12px; margin-bottom: 16px;"> <div style="background: #ecfdf5; padding: 10px; border-radius:
---	--



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```

    <?php if ($selection['status'] ===
    style="display:none; margin: 24px;

    'Preparing'): ?>
    padding: 20px; background: #f8fafc;

    <button
    border: 1px solid #e2e8f0; border-radius:

    onclick="document.getElementById('ad
    12px;">

    dPositionForm').style.display='block'"
    <form

    style="background: #4f46e5; color:
    action="api/add_position.php"

    white; border: none; padding: 8px 16px;
    method="POST">

    border-radius: 8px; font-size: 0.875rem;
    <input type="hidden"

    font-weight: 600; cursor: pointer;
    name="election_id" value="<?=

    display: flex; align-items: center; gap:
    $selection['id'] ?>">

    6px;">
    <div style="display: grid;

    <i data-lucide="plus"
    grid-template-columns: 2fr 1fr auto; gap:

    style="width: 14px;"></i>
    16px; align-items: flex-end;">

    Add Position
    <div>

    </button>
    <label

    <?php endif; ?>
    style="display:block; font-size: 0.75rem;

    </div>
    font-weight: 600; color: #64748b;

    margin-bottom: 6px;">Position

    <div id="addPositionForm"
    Name</label>
  
```



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```
<input type="text"
name="name" style="width: 100%;
padding: 10px; border: 1px solid
#e2e8f0; border-radius: 8px; outline:
none;" placeholder="e.g. President"
required>

</div>

<div>
  <label
style="display: block; font-size: 0.75rem;
font-weight: 600; color: #64748b;
margin-bottom: 6px;">Max
Votes</label>

  <input type="number"
name="max_votes" style="width: 100%;
padding: 10px; border: 1px solid
#e2e8f0; border-radius: 8px; outline:
none;" value="1" min="1">

</div>

<div style="display: flex;
gap: 8px;">
  <button type="submit"
style="background: #2563eb; color:
white; border: none; padding: 10px
20px; border-radius: 8px; font-weight:
600; cursor: pointer;">Save</button>

  <button type="button"
onclick="document.getElementById('ad
dPositionForm').style.display='none'"
style="background: #f1f5f9; color:
#475569; border: none; padding: 10px
20px; border-radius: 8px; font-weight:
600; cursor: pointer;">Cancel</button>
</div>
</form>
</div>
```



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```
<div style="overflow-x: auto;">                                font-weight: 700; color: #64748b;

<table style="width: 100%;                                text-transform:
border-collapse: collapse;">                                uppercase;">Candidates</th>

    <thead>                                                    <th style="padding: 12px
        <tr style="background:                                24px; text-align: right; font-size:
#f9fafb; border-bottom: 1px solid                            0.75rem; font-weight: 700; color:
#f3f4f6;">                                                    #64748b; text-transform:
        <th style="padding: 12px                                uppercase;">Actions</th>
24px; text-align: left; font-size: 0.75rem;                    </tr>
font-weight: 700; color: #64748b;                                </thead>
text-transform: uppercase;">Position                            <tbody>
Name</th>                                                        <?php if (empty($positions)):

    <th style="padding: 12px                                ?>
24px; text-align: left; font-size: 0.75rem;                    <tr>
font-weight: 700; color: #64748b;                                <td colspan="4"
text-transform: uppercase;">Max                                style="padding: 48px; text-align: center;
Votes</th>                                                        color: #94a3b8;">

    <th style="padding: 12px                                <div style="display: flex;
24px; text-align: left; font-size: 0.75rem;                    flex-direction: column; align-items:
```



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center; gap: 12px;">

\$cStmt->fetchColumn();

<i data-lucide="layers"

?>

style="width: 48px; height: 48px; color:

<tr style="border-bottom:

#e2e8f0;"></i>

1px solid #f3f4f6; transition: background

No positions

0.2s;"

defined yet. Start by adding one.

onmouseover="this.style.background='f

</div>

8fafc"

</td>

onmouseout="this.style.background='tra

</tr>

nsparent">

<?php else: ?>

<td style="padding: 16px

<?php foreach (\$positions as

24px; font-weight: 600; color:

\$pos):

#1e293b;"><?=

\$cStmt =

htmlspecialchars(\$pos['name']) ?></td>

\$pdo->prepare("SELECT COUNT(*)

<td style="padding: 16px

FROM candidates WHERE position_id

24px; color: #4b5563;">

= ?");

<span

style="background: #f1f5f9; padding:

\$cStmt->execute([\$pos['id']]);

4px 10px; border-radius: 6px; font-size:

\$cCount =

0.75rem; font-weight: 600;"><?=



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```
$pos['max_votes'] ?> Vote<?=
                                id=<?= $pos['id'] ?>" style="color:
                                #4f46e5; text-decoration: none;
                                font-size: 0.875rem; font-weight: 600;
                                display: inline-flex; align-items: center;
                                gap: 6px;">
                                Manage Candidates
                                <i
                                data-lucide="arrow-right" style="width:
                                height: 8px; border-radius: 50%;
                                14px;"></i>
                                background: #10b981;"></div>
                                </a>
                                <span
                                </td>
                                style="font-weight: 600; color:
                                </tr>
                                #1e293b;"><?= $cCount ?></span>
                                <?php endforeach; ?>
                                </div>
                                <?php endif; ?>
                                </td>
                                </tbody>
                                <td style="padding: 16px
                                </table>
                                24px; text-align: right;">
                                </div>
                                <a
                                </div>
                                href="manage_candidates.php?position_
```



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```
<div style="display: flex;                letter-spacing: 0.05em;">Total
flex-direction: column; gap: 24px;">  Candidates</div>

<div style="background: #ffffff;          <?php
border: 1px solid #f3f4f6; border-radius:    $scanStmt =
16px; padding: 24px;">                    $pdo->prepare("SELECT COUNT(*)
<h2 style="margin-top: 0;                  FROM candidates WHERE election_id
font-size: 1.1rem; color: #1e293b;         = ?");
font-weight: 700; margin-bottom:           $scanStmt->execute([$id]);
20px;">Quick Stats</h2>                  $candidatesTotal =
                                           $scanStmt->fetchColumn();

<div style="display: flex;                  ?>
flex-direction: column; gap: 20px;">      <div style="font-size:
                                           1.5rem; font-weight: 800; color:
                                           #1e293b;"><?= $candidatesTotal
                                           ?></div>

<div>                                     </div>

<div style="font-size:                    <div style="background:
0.7rem; color: #64748b; text-transform:    #fef2f2; padding: 12px; border-radius:
uppercase; font-weight: 700;              12px; color: #ef4444;">
```



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```
<i                                #f0fdf4; padding: 12px; border-radius:
data-lucide="user-check"></i>    12px; color: #22c55e;">
</div>                            <i>
</div>                            data-lucide="bar-chart-3"></i>
                                </div>
<div style="display: flex;        </div>
justify-content: space-between;  </div>
align-items: center;">        </div>
<div>
    <div style="font-size:        <div style="background:
0.7rem; color: #64748b; text-transform:    linear-gradient(135deg, #4f46e5 0%,
uppercase; font-weight: 700;              #3730a3 100%); border-radius: 16px;
letter-spacing: 0.05em;">Voter          padding: 24px; color: white;">
Turnout</div>                            <h3 style="margin: 0; font-size:
    <div style="font-size:          1rem; font-weight: 700; margin-bottom:
1.5rem; font-weight: 800; color:          12px;">Need Help?</h3>
#1e293b;"><?= $votesCount ?></div>    <p style="font-size: 0.875rem;
</div>                                opacity: 0.9; margin-bottom:
<div style="background:            20px;">Manage your election structure
```



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by adding positions and assigning
 candidates to them. Once ready, you can
 launch the election.</p>

style="display: block; width: 100%;
 text-align: center; background: white;
 color: #4f46e5; padding: 12px;
 border-radius: 8px; font-weight: 700;
 text-decoration: none; font-size:
 0.875rem;">View Reports

</div>

</div>

</div>

</main>

</div>

<div id="editElectionModal"

class="modal">

<div class="modal-content

animate-scale-in" style="max-width:
 550px;">

<div class="modal-header">

<h2 style="font-weight: 800;">Edit

Election Details</h2>

<button type="button"

onclick="document.getElementById('edi
 tElectionModal').style.display='none'"

class="close-btn">×</button>

</div>

<form

action="api/update_election.php"

method="POST">

<input type="hidden" name="id"

value="<?= \$selection['id'] ?>">

<div class="form-group"

style="margin-bottom: 20px;">

<label>Election Title</label>

<input type="text" name="title"



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<pre>value="<?=" 20px; margin-bottom: 32px;"> htmlspecialchars(\$selection['title']) ?>" style="width: 100%; padding: 12px; border: 1px solid #e2e8f0; border-radius: 10px; outline: none;" required> </div> <div class="form-group" style="margin-bottom: 20px;"> <label>Description</label> <textarea name="description" rows="3" style="width: 100%; padding: 12px; border: 1px solid #e2e8f0; border-radius: 10px; outline: none; font-family: inherit;"><?=" htmlspecialchars(\$selection['description'])) ?></textarea> </div> <div style="display: grid; grid-template-columns: 1fr 1fr; gap:</pre>	<pre>> <div class="form-group"> <label>Start Date & Time</label> <input type="datetime-local" name="start_date" value="<?=" date('Y-m-d\TH:i', strtotime(\$selection['start_date_and_time'])) ?>" style="width: 100%; padding: 12px; border: 1px solid #e2e8f0; border-radius: 10px; outline: none;" required> </div> <div class="form-group"> <label>End Date & Time</label> <input type="datetime-local" name="end_date" value="<?=" date('Y-m-d\TH:i',</pre>
--	--



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```
strtotime($selection['end_date_and_time'])

)) ?>" style="width: 100%; padding:
12px; border: 1px solid #e2e8f0;
border-radius: 10px; outline: none;"
required>

</div>

</div>

<div class="modal-footer">

<button type="button"

onclick="document.getElementById('edi
tElectionModal').style.display='none'"
class="btn-cancel">Cancel</button>

<button type="submit"

class="btn-submit" style="background:
#4f46e5;">Save Changes</button>

</div>

</form>

</div>

</div>
```

```
<script>

    lucide.createIcons();

    window.onclick = function(event) {

        if (event.target.id ===

            'editElectionModal') {

                event.target.style.display =

                    'none';

            }

        }

    }

</script>

</body>

</html>

<?php

declare(strict_types=1);

require_once __DIR__ .

    '/auth_helper.php';

require_login();

$user = get_user();
```



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```
// Check if already voted

$id = $_GET['id'] ?? '';

if (!$id) {

    header("Location: index.php");

    exit;

}

$stmt = $pdo->prepare("SELECT *
FROM elections WHERE id = ?");

$stmt->execute([$id]);

$selection = $stmt->fetch();

if (!$selection || $selection['status'] !==
'Ongoing') {

    die("Election is not currently
ongoing.");

}

$check = $pdo->prepare("SELECT
COUNT(*) FROM votes WHERE
election_id = ? AND voter_id = ?");

$check->execute([$id, $user['id']]);

if ($check->fetchColumn() > 0) {

    header("Location:
index.php?error=already_voted");

    exit;

}

$positions = $pdo->prepare("SELECT *
FROM positions WHERE election_id =
? ORDER BY sort_order ASC");

$positions->execute([$id]);

$positions = $positions->fetchAll();

$endTime =

strtotime($selection['end_date_and_time'])
```



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```
) * 1000;                                </script>

?>                                         <style>

<!doctype html>                           :root {

<html lang="en">                           --primary: #4f46e5;

<head>                                    --primary-hover: #4338ca;

  <meta charset="utf-8" />                --primary-light: #eef2ff;

  <meta name="viewport"                   --bg: #f8fafc;

  content="width=device-width,            --text: #1e293b;

  initial-scale=1" />                    --text-muted: #64748b;

  <title>Ballot: <?=                      --border: #e2e8f0;

  htmlspecialchars($selection['title'])    }

?></title>

  <link                                   body {

  href="https://fonts.googleapis.com/css2?  background-color: var(--bg);

  family=Plus+Jakarta+Sans:wght@400;5    color: var(--text);

  00;600;700;800&display=swap"           font-family: 'Plus Jakarta Sans',

  rel="stylesheet">                       sans-serif;

  <script                                 margin: 0;

  src="https://unpkg.com/lucide@latest">  line-height: 1.5;
```



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```
}
text-decoration: none;

letter-spacing: -0.025em;

.navbar {
background: white;

padding: 1rem 2rem;

display: flex;

justify-content: space-between;

align-items: center;

border-bottom: 1px solid

}

var(--border);

position: sticky;

top: 0;

z-index: 1000;

}

.ballot-container {
max-width: 900px;

margin: 0 auto;

padding: 40px 20px 120px;

}

.ballot-title-area {
margin-bottom: 40px;

}

.ballot-title-area h1 {
font-size: 1.875rem;

font-weight: 700;

margin: 0 0 8px 0;

color: #1e293b;

}

.brand {
font-weight: 800;

font-size: 1.25rem;

color: var(--primary);

}
```



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```
} font-size: 1rem;

font-weight: 600;

.ballot-title-area p { color: #475569;
                        margin-bottom: 16px;
                        margin: 0; }
                        font-size: 1rem;
}

.voter-info-grid {
    display: grid;
    grid-template-columns: repeat(4,
                                1fr);
    gap: 20px;
    border: 1px solid var(--border);
    border-radius: 12px;
    padding: 24px;
    margin-bottom: 32px;
    box-shadow: 0 1px 3px
                rgba(0,0,0,0.05);
}

.voter-info-card {
    background: white;
    border: 1px solid var(--border);
    border-radius: 12px;
    padding: 24px;
    margin-bottom: 32px;
    box-shadow: 0 1px 3px
                rgba(0,0,0,0.05);
}

.voter-info-title {
    text-transform: uppercase;
```



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```
margin-bottom: 4px;
border: 1px solid #fee2e2;
}

.info-item span {
  font-size: 0.875rem;
  font-weight: 600;
  color: #1e293b;
}

.position-group {
  background: white;
  border: 1px solid var(--border);
  border-radius: 12px;
  padding: 32px;
  margin-bottom: 24px;
  box-shadow: 0 1px 3px
    rgba(0,0,0,0.05);
}

.timer-container {
  display: flex;
  align-items: center;
  gap: 8px;
  color: #ef4444;
  font-weight: 700;
  font-size: 0.875rem;
  background: #fef2f2;
  padding: 8px 16px;
  border-radius: 8px;
}

.position-title {
  font-size: 1.25rem;
  font-weight: 700;
  color: #1e293b;
  margin-bottom: 24px;
}
```



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```
gap: 16px;

.candidates-list {
  background: white;
  display: flex;
}

flex-direction: column;

gap: 12px;
}

.candidate-card:hover {
  border-color: var(--primary);
  background: #fcfdff;
  transform: translateY(-2px);
  box-shadow: 0 10px 15px -3px
    rgba(0,0,0,0.1);
}

.candidate-label {
  cursor: pointer;
  display: block;
}

.candidate-card {
  border: 1px solid var(--border);
  border-radius: 12px;
  padding: 16px 20px;
  transition: all 0.2s;
  display: flex;
  align-items: center;

  input[type="radio"]:checked +
    .candidate-card {
      border-color: var(--primary);
      background: white;
      box-shadow: 0 0 0 1px
        var(--primary);
    }
}
```



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```
content: "";

.radio-circle {
  width: 20px;
  height: 20px;
  border: 2px solid #cbd5e1;
  border-radius: 50%;
  display: flex;
  align-items: center;
  justify-content: center;
  transition: all 0.2s;
}

input[type="radio"]:checked +
.candidate-card .radio-circle {
  border-color: var(--primary);
}

input[type="radio"]:checked +
.candidate-card .radio-circle::after {
  content: "";
  width: 10px;
  height: 10px;
  background: var(--primary);
  border-radius: 50%;
}

.candidate-avatar {
  width: 48px;
  height: 48px;
  background: #f1f5f9;
  border-radius: 50%;
  overflow: hidden;
  display: flex;
  align-items: center;
  justify-content: center;
  font-weight: 700;
  color: #64748b;
}
```



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```
.candidate-avatar img {  
    width: 100%;  
    height: 100%;  
    object-fit: cover;  
}  
  
.candidate-info h3 {  
    margin: 0;  
    font-size: 1rem;  
    font-weight: 600;  
    color: #1e293b;  
}  
  
.candidate-info p {  
    margin: 2px 0 0 0;  
    font-size: 0.875rem;  
    color: #64748b;  
}  
  
.submit-bar {  
    position: fixed;  
    bottom: 0;  
    left: 0;  
    right: 0;  
    background: white;  
    padding: 16px 40px;  
    border-top: 1px solid var(--border);  
    display: flex;  
    justify-content: center;  
    z-index: 100;  
    box-shadow: 0 -10px 15px -3px  
        rgba(0,0,0,0.05);  
}  
  
.submit-container {  
    max-width: 900px;  
    width: 100%;
```



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```
display: flex;
}

justify-content: space-between;

align-items: center;
}

.btn-submit {
background: var(--primary);
color: white;
border: none;
padding: 14px 40px;
border-radius: 12px;
font-size: 1rem;
font-weight: 700;
cursor: pointer;
transition: all 0.2s;
box-shadow: 0 4px 6px -1px
    rgba(79, 70, 229, 0.2);
}

.submit-text h4 {
margin: 0;
font-size: 1.25rem;
font-weight: 700;
color: var(--text);
}

.btn-submit:hover {
background: var(--primary-hover);
transform: translateY(-1px);
}
```



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```
        box-shadow: 0 10px 15px -3px
        rgba(79, 70, 229, 0.3);
    }

    align-items: center;
    justify-content: center;
    padding: 20px;
}

.btn-submit:active {
    transform: translateY(0);
}

.modal-overlay.active {
    display: flex;
}

/* Modal Styles */

.modal-overlay {
    display: none;
    position: fixed;
    top: 0;
    left: 0;
    right: 0;
    bottom: 0;
    background: rgba(15, 23, 42, 0.6);
    backdrop-filter: blur(4px);
    z-index: 2000;
}

.confirm-modal {
    background: white;
    border-radius: 20px;
    max-width: 450px;
    width: 100%;
    padding: 32px;
    box-shadow: 0 25px 50px -12px
        rgba(0, 0, 0, 0.25);
    animation: modalIn 0.3s
        cubic-bezier(0.34, 1.56, 0.64, 1);
}
```



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}

}

```
@keyframes modalIn {  
  from { transform: scale(0.9)  
    translateY(20px); opacity: 0; }  
  to { transform: scale(1)  
    translateY(0); opacity: 1; }  
}
```

```
.confirm-modal h2 {  
  margin: 0 0 12px 0;  
  font-size: 1.5rem;  
  font-weight: 800;  
  color: #1e293b;  
}
```

```
.modal-icon {  
  width: 64px;  
  height: 64px;  
  background: #eef2ff;  
  color: #4f46e5;  
  border-radius: 16px;  
  display: flex;  
  align-items: center;  
  justify-content: center;  
  margin-bottom: 24px;
```

```
.confirm-modal p {  
  margin: 0 0 32px 0;  
  color: #64748b;  
  font-size: 1rem;  
  line-height: 1.6;  
}
```

```
.modal-actions {  
  display: grid;  
  grid-template-columns: 1fr 1fr;
```



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```
gap: 16px;

}

.btn-modal {

padding: 12px 24px;

border-radius: 12px;

font-size: 0.875rem;

font-weight: 700;

cursor: pointer;

transition: all 0.2s;

border: none;

text-align: center;

}

.btn-modal-cancel {

background: #f1f5f9;

color: #64748b;

}

.btn-modal-cancel:hover {

background: #e2e8f0;

color: #1e293b;

}

.btn-modal-confirm {

background: #4f46e5;

color: white;

box-shadow: 0 4px 6px -1px

rgba(79, 70, 229, 0.2);

}

.btn-modal-confirm:hover {

background: #4338ca;

box-shadow: 0 10px 15px -3px

rgba(79, 70, 229, 0.3);

}

input[type="radio"] { display: none; }
```



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```

    <a href="logout.php" style="color:
    #ef4444; font-size: 0.875rem;
    font-weight: 700; text-decoration:
    none;">Logout</a>
  </div>
</nav>
</style>
</head>
<body class="<?=( $user['theme'] ??
'light') === 'dark' ? 'dark-theme' : " ?>">
  <nav class="navbar">
    <a href="index.php"
    class="brand">Click to Vote</a>
    <div>
      <span style="margin-right: 1rem;
      color: var(--text-muted); font-size:
      0.875rem; font-weight: 600;"><?=(
      htmlspecialchars($user['name'])
      ?></span>
    </div>
    <div class="ballot-container">
      <div class="ballot-title-area">
        <h1><?=(
        htmlspecialchars($selection['title'])
        ?></h1>
        <p>Review the candidates carefully
        and cast your secure vote below.</p>
      </div>
      <div class="voter-info-card">
        <div style="display: flex;
        justify-content: space-between;

```



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align-items: flex-start; margin-bottom: 20px;"> <div class="voter-info-title">Voter Information</div> <div class="timer-container"> <i data-lucide="clock" style="width: 16px;"></i> 00:00:00 </div> </div> <div class="voter-info-grid"> <div class="info-item"> <label>Email</label> <?= htmlspecialchars(\$user['email']) ?> </div>	<div class="info-item"> <label>Grade Level</label> Grade <?= htmlspecialchars((string)(\$user['grade_le vel'] ?? 'N/A')) ?> </div> <div class="info-item"> <label>Track/Cluster</label> <?= htmlspecialchars(\$user['track'] ?? 'N/A') ?> </div> <div class="info-item"> <label>Section</label> <?= htmlspecialchars(\$user['section'] ?? 'N/A') ?> </div> </div>
--	---



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```

</div>
                                u.name, u.track FROM candidates c

                                JOIN users u ON c.user_id = u.id

                                WHERE c.position_id = ? AND

                                c.approved = TRUE";

                                $params = [$pos['id']];

                                <input type="hidden"

                                name="election_id" value="<?= $id                                if ($pos['type'] === 'Track

                                ?>">                                Specific') {

                                $sql .= " AND u.track =

                                <?php foreach ($positions as $index                                ?";

                                => $pos): ?>                                $params[] =

                                <div class="position-group">                                $user['track'];

                                <div class="position-title">                                }

                                <?=

                                htmlspecialchars($pos['name']) ?>                                $cStmt =

                                </div>                                $pdo->prepare($sql);

                                $cStmt->execute($params);

                                <?php                                $candidates =

                                $sql = "SELECT c.*,                                $cStmt->fetchAll();
  
```



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```
?> class="candidate-label">

                                <input type="radio"

<?php if (empty($candidates)): name="votes[<?=$pos['id'] ?>]"

?> value="<?=$cand['id'] ?>" required>

                                <div

                                <div

background: #f8f9fc; border-radius: class="candidate-card">

12px; text-align: center; border: 1px                                <div

dashed #cbd5e1;">                                class="radio-circle"></div>

                                <div

                                <div

color: #64748b; font-size:                                class="candidate-avatar">

0.875rem;">No candidates available for                                <?=

your track.</p>                                substr($cand['name'], 0, 1) ?>

                                </div>                                </div>

                                <div

                                <div

                                <div

                                class="candidate-info">

class="candidates-list">                                <h3><?=

                                <?php foreach                                htmlspecialchars($cand['name'])

($candidates as $cand): ?>                                ?></h3>

                                <label                                <p><?=
```



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<pre> htmlspecialchars(\$cand['party_name'] ? 'Independent') ?></p> </div> </div> </label> <?php endforeach; ?> </div> <?php endif; ?> </div> <?php endforeach; ?> <div class="submit-bar"> <div class="submit-container"> <div class="submit-text"> <p>Ready to submit?</p> <h4>Review your selections</h4> </div> <button type="button" </pre>	<pre> class="btn-submit" onclick="openConfirmModal()"> Cast My Vote </button> </div> </div> </div> </form> </div> <!-- Confirmation Modal --> <div id="confirmModal" class="modal-overlay"> <div class="confirm-modal"> <div class="modal-icon"> <i data-lucide="shield-check" style="width: 32px; height: 32px;"></i> </div> <h2>Cast your vote?</h2> <p>You are about to submit your </pre>
---	--



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choices. This action is permanent and

```
var(--border);">
```

cannot be undone. Are you sure you

```
&copy; <?= date('Y') ?> Click to Vote
```

want to proceed?</p>

| High School Online Election System

```
<div class="modal-actions">
```

```
</footer>
```

```
<button type="button"
```

```
class="btn-modal btn-modal-cancel"
```

```
<script>
```

```
onclick="closeConfirmModal()">Revie
```

```
lucide.createIcons();
```

```
w Choices</button>
```

```
<button type="button"
```

```
function openConfirmModal() {
```

```
class="btn-modal btn-modal-confirm"
```

```
// Basic validation: check if all
```

```
onclick="submitBallot()">Yes, Cast My
```

```
required radios are checked
```

```
Vote</button>
```

```
const form =
```

```
</div>
```

```
document.getElementById('ballotForm');
```

```
</div>
```

```
if (!form.checkValidity()) {
```

```
</div>
```

```
form.reportValidity();
```

```
return;
```

```
<footer style="text-align: center; color:
```

```
}
```

```
var(--text-muted); padding: 4rem 2rem;
```

```
background: white; border-top: 1px solid
```

```
document.getElementById('confirmMod
```



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```
al').classList.add('active');  
  
}
```

```
const now = new Date().getTime();  
  
const distance = endTime - now;
```

```
function closeConfirmModal() {
```

```
if (distance < 0) {
```

```
document.getElementById('confirmMod  
al').classList.remove('active');  
  
}
```

```
document.getElementById("countdown"  
).innerHTML = "EXPIRED";
```

```
function submitBallot() {
```

```
document.getElementById("ballotForm"  
).style.opacity = "0.5";
```

```
document.getElementById('ballotForm').  
submit();  
  
}
```

```
document.getElementById("ballotForm"  
).style.pointerEvents = "none";  
  
return;  
  
}
```

```
// Countdown Timer
```

```
const endTime = <?= $endTime ?>;
```

```
const hours = Math.floor((distance  
% (1000 * 60 * 60 * 24)) / (1000 * 60 *  
60));
```

```
function updateCountdown() {
```



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```

const minutes =
Math.floor((distance % (1000 * 60 *
60)) / (1000 * 60));
const seconds =
Math.floor((distance % (1000 * 60)) /
1000);

setInterval(updateCountdown, 1000);
document.getElementById("countdown"
).innerHTML =
(hours < 10 ? "0" : "") + hours +

```



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APPENDIX S

Documentation

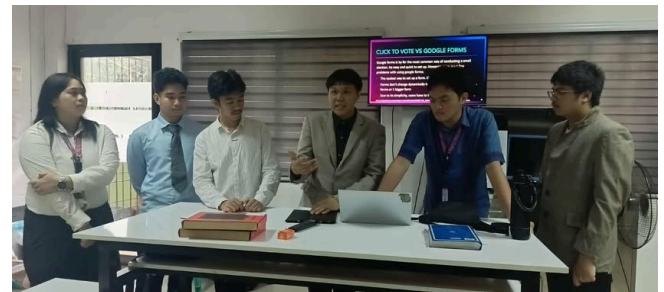
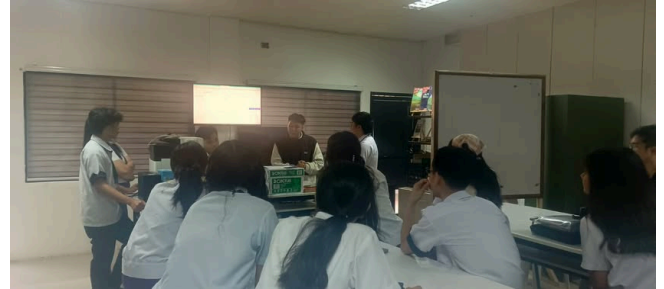


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APPENDIX T

Grammar Certificate



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CERTIFICATION

March, 2026

This is to certify that I have checked and edited the manuscript submitted by **Bugayong Rezzo, Castro Zuelyka, Palomar Xian Michael, Santiago Jay Orly Mil, Tse Wing Kurt Leovince, Ymbang Clark Rhazen**, Senior High School learners, entitled **"Click to Vote: A Digital Voting System for a Public Senior High School"** and found it to be:

_____ Grammatically correct
_____ Properly edited
_____ Ready for submission
_____ With minor revisions required
_____ With major revisions required

Signed this ____ day of _____ 2026 at _____

IRENE AYAG TORREFRANCA
STVEP English Teacher
Iloilo National High School



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APPENDIX U

Curriculum Vitae



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QUALIFICATION

- Knowledgeable in basic Programming and HTML
- Has knowledge in writing and editing

EDUCATIONAL BACKGROUND

Senior High School

Iloilo National High School, (2024-2026)

Junior High School

Iloilo National High School, (2020-2024)

Elementary

Progressive Learning Activities for the Young Integrated School, (2014-2020)

AWARDS AND RECOGNITION

- Contributed to group projects and activities aligned with ICT



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QUALIFICATION

- Basic Knowledge in Software Applications
- Knowledgeable in basic Programming and HTML
- Has knowledge in writing and editing

EDUCATIONAL BACKGROUND

Senior High School

Iloilo National High School, (2024-2026)

Junior High School

Iloilo National High School, (2020-2024)

Elementary

La Paz II Elementary School, (2015-2020)

AWARDS AND RECOGNITION

- Contributed to group projects and activities aligned with ICT and sustainability goals



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Mobile No: 0985-092-1741



QUALIFICATION

- Quick decision maker
- Active listener
- Technology Enthusiast
- Skills in programming specifically in HTML and JAVA.

EDUCATIONAL BACKGROUND

Senior High School

Iloilo National High School, (2024-2026)

Junior High School

Iloilo National High School, (2020-2024)

Elementary

Tapaz Elementary School, (2017-2020)

AWARDS AND RECOGNITION

- Elementary- with honors
- Highschool- with honors



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Mobile No: 0919-706-1648



QUALIFICATION

- Experienced in many programming languages which include:
 - a. Python (Pygame, React, Django)
 - b. Java
 - c. C# and C++
 - d. HTML and its support languages (JS, CSS, PHP)
- Has hands on experience with coding since grade 5 with python being the starting language

EDUCATIONAL BACKGROUND

Senior High School

Iloilo National High School (2024-2026)

Junior High School

Iloilo National High School (2020-2024)

Elementary

Baptist Center Christian School of Iloilo City Inc (2011-2020)

AWARDS AND RECOGNITION

- W/ Honors and W/ High Honors during JHS to SHS
- 1st place during INHS' first programming competition (SY 2024-2025)
- Nominated for Best in Programming and Best in Immersion



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Iloilo City

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QUALIFICATION

- Basic knowledge in Software Applications
- Knowledgeable in basic Java Programming Language and HTML
- Has knowledge in basic editing

EDUCATIONAL BACKGROUND

Senior High School

Iloilo National High School, (2024-2026)

Junior High School

Leganes National High School, (2020-2024)

Elementary

Napnud Elementary School, (2017-2020)

AWARDS AND RECOGNITION

- With Honors
- 2nd Place winner of Essay Writing in Elementary



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Mobile No: 0950-891-0331



QUALIFICATION

- **Languages:** Java, JavaScript, HTML, CSS, PHP
- **Software:** VS Code, NetBeans, Github, XAMPP, FileZilla, Microsoft Office, Google Workspace, Canva
- **Interest:** Work well under pressure, analytical skills, problem-solving skills, communication skills, basic troubleshooting

EDUCATIONAL BACKGROUND

Senior High School

Iloilo National High School, (2024 - 2026)

Junior High School

Philippine American Progress School, Inc., (2020 - 2024)

Elementary

Philippine American Progress School, Inc., (2013-2020)

AWARDS AND RECOGNITION

- Elementary: With Honors (Grade 4, Grade 5, Grade 6)
- Junior High School: With Honors (Grade 9, Grade 10)
- PAPS's 26th Founding Anniversary Quiz Bee Competition 2nd Placer
- 18th National Scout Jamboree Participant



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